



ATLAS Note

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ATLAS simulation of boson plus jets processes in Run 2

The ATLAS Collaboration

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This note describes the ATLAS simulation setup used to model the production of single electroweak bosons (W , Z/γ^* and prompt γ) in association with jets in proton–proton collisions at centre-of-mass energies of 8 and 13 TeV. Several Monte Carlo generator predictions are compared in regions of phase space relevant for data analyses during the LHC Run-2, or compared to unfolded data distributions measured in previous Run-1 or early Run-2 ATLAS analyses. Comparisons are made for regions of phase space with or without additional requirements on the heavy-flavour content of the accompanying jets, as well as electroweak Vjj production processes. Both higher-order corrections and systematic uncertainties are also discussed.



1 Introduction

The production of electroweak vector bosons (W and Z/γ^* , which will be indicated in the remainder of this document as V) is a very important process at the LHC. Measurements of such processes allow to gain a deeper understanding of pQCD and to test the currently available predictions. These analyses are generally carried out looking at final states containing electrons or muons, exploiting relatively clean experimental signatures. With the higher centre-of-mass energy as well as the increased luminosity expected for the LHC Run 2, rarer processes like the electroweak production of vector bosons in association with two jets, including vector boson fusion processes, become more accessible as well.

Furthermore, V + jets production is also a major background to a large number of analyses, such as top-quark, Higgs-boson measurements, and searches for new physics phenomena. An accurate modelling of the data is therefore necessary in this type of analyses, and often an accuracy beyond the calculated order is desired.

Typical Monte Carlo (MC) generators that are used to generate V + jets events combine multi-leg matrix elements at LO [1–3] or even NLO [4] with the parton shower in order to include additional parton emissions at matrix-element accuracy. Merging algorithms are used to avoid double counting the final-state configurations that are also produced by the parton shower. It is clear that the production of reliable and high-statistics Monte Carlo (MC) generator samples is a key ingredient at multi-purpose hadron collider experiments. In the following sections, the main generators used in the ATLAS experiment [5] during Run 2 will be discussed and studied in detail.

In this note, comparisons of different generator predictions for V + jets processes are made at centre-of-mass energies of 8 and 13 TeV, including comparisons to recent ATLAS measurements. The studies presented follow naturally from the ones presented in [6], which focussed on comparisons in regions of phase space used mainly in Run 1 analyses. For this note, comparisons are made in regions of phase space more suitable for Run 2 analyses, with a particular focus on the production of vector bosons in association with heavy-flavour jets (also in the context of the search for Higgs decays to a b -quark pair in the same final state) and a study of systematic uncertainties for electroweak Vjj production is also performed. The scope of this note is twofold; the aim is to document the baseline samples used by ATLAS analyses in Run 2, whilst also providing insight into the differences between these samples in regions of phase space relevant to the LHC analyses, including the size of systematic uncertainties on the theoretical modelling. In addition, preliminary studies on a specific MG5_aMC+Py8 CKKW-L tune, with rapidity-ordering shower are compared to simpler PyTHIA8-only tunes.

Throughout this note, V + jets predictions are only shown for the $Z/\gamma^* \rightarrow e^+e^-$ and $W^\pm \rightarrow e^\pm\nu$ channel (with the exception of the study dedicated to the $VH(H \rightarrow b\bar{b})$ background studies in Section 5.3) and the ATLAS coordinate system is adopted, which consists of a right-handed system with the z -direction defined by the beam direction, and x pointing towards the centre of the LHC ring. The polar angle θ is defined relative to the z axis, and pseudorapidity $\eta = -\ln(\tan(\theta/2))$, with the transverse direction corresponding to $\theta = \pi/2$, $\eta = 0$. Transverse momentum, p_T , is defined as the component of momentum in the transverse direction. The polar angle, ϕ , is defined in the $x - y$ plane. Angular separation is defined by $\Delta R = \sqrt{\Delta\phi^2 + \Delta\eta^2}$. Rapidity is defined by $y = 0.5 \times (E + p_z)/(E - p_z)$, where E is the total energy, and p_z the component of the momentum in the z direction.

2 Baseline generators

In the following paragraphs, the setup of the samples used in this note for the different set of comparisons is described. SHERPA and MG5_aMC@NLO are the two most commonly used $V + \text{jets}$ generators for ATLAS Run 2 analyses using the 2015 and 2016 datasets. The focus is towards the usage of modern generator predictions, in particular the development of a NLO-merged setups.

2.1 SHERPA

SHERPA [7] is a parton shower Monte Carlo generator simulating additional hard parton emissions [8] that are matched to a parton shower based on Catani–Seymour subtraction terms [9]. The merging of multi-parton matrix elements with the parton shower is achieved using an improved CKKW matching procedure [1, 2], which is extended to next-to-leading order (NLO) accuracy using the MEPS@NLO prescription [4]. Virtual corrections for matrix elements at NLO accuracy are provided by OPENLOOPS [10] using the COLLIER tensor integral reduction library [11]. SHERPA v2.2 is used as a baseline using parton density functions (PDFs) provided by the NNPDF3.0nnlo set [12] as well as the dedicated set of tuned parton-shower parameters developed by the SHERPA authors for this version.

Matrix elements for the $\ell^+\ell^- + \text{jets}$ and $\ell\nu + \text{jets}$ final states (with $\ell = e, \mu, \tau$) have been generated using SHERPA v2.2 with up to two additional parton emissions at NLO accuracy and up to four additional parton emissions at LO accuracy. The matrix elements are merged with the SHERPA parton shower using the MEPS@NLO prescription using the five-flavour-number scheme (5FNS) with a matching scale of 20 GeV. The invariant mass of the dilepton system for the $\ell^+\ell^- + \text{jets}$ final states is required to be at least 40 GeV.

Each final state is sliced in the variable $\max[H_T, p_T(V)]$ at the parton level, where the boson transverse momentum, $p_T(V)$, is worked out from the two parton-level leptons and the parton-level H_T is given by the scalar p_T sum of all parton-level jets with $p_T > 20$ GeV. Here, parton-level jets are constructed from all remaining matrix-element-level partons (excluding the two leptons) using the anti- k_t algorithm [13] with a jet-radius parameter 0.4. Below $\max[H_T, p_T(V)]$ 500 GeV, the slices are also filtered according to their b -hadron and c -hadron content at the particle level.

Separate samples are generated for electroweak Vjj production, with two additional electroweak couplings at tree level. Matrix elements for the $\ell^+\ell^-jj$ and $\ell\nu jj$ final states have been generated using SHERPA v2.2 with up to two additional parton emissions at LO accuracy beyond the first two jets. At tree level, this gives rise to a gauge-invariant set of seven Vjj diagrams that involve a t -channel exchange of a heavy gauge boson using three electroweak couplings (and an additional one for the decay of the boson). The matrix elements are merged with the SHERPA parton shower using the MEPS@LO prescription with a matching scale of 15 GeV. The invariant mass of the dilepton system for the $\ell^+\ell^-jj$ final states is required to be at least 40 GeV.

Contributions from top-quark interactions are avoided by generating matrix elements without bottom quarks for the hard-scattering process. The `Min_N_TChannels` option has been activated in order to remove contributions from diboson interactions that do not contain a t -channel exchange of a heavy gauge boson. The Vjj samples are produced using the G_μ scheme, making use of the tree-level relations between the electroweak input parameters.

Although not explicitly shown in this note, samples for single photon production in association with jets are produced using the SHERPA v2.2 generator. Matrix elements are calculated for up to two jets at NLO accuracy and up to four jets at LO accuracy and merged with the SHERPA parton shower using the MEPS@NLO prescription. The merging scale has been set up dynamically in the scheme of [14] using $\bar{Q}_{\text{cut}} = 20 \text{ GeV}$. Further phase space cuts are applied at the parton level to require $p_T^\gamma > 70 \text{ GeV}$ and a smooth hadronic isolation [15] of the photon with $n = 2$, $\epsilon = 0.1$, $\delta = 0.1$. The QCD factorisation and renormalisation scales were set using the STRICT_METS scale setter with a core scale of $\mu = p_\perp^\gamma$, while α_{QED} is evaluated in the Thomson limit.

All SHERPA 2.2 samples mentioned above have been generated with additional weights for the NNPDF3.0nnlo replicas, α_S variations of ± 0.002 around the nominal value of 0.118, as well as 7-point scale variations of the factorisation and renormalisation scales.

2.2 MadGRAPH5_aMC@NLO using CKKW-L

Matrix elements for $V +$ up to 4 partons at LO accuracy are produced using MadGRAPH5_aMC@NLO v2.2.2 [16] interfaced with Pythia v8.186 [17] for the modelling of the parton shower and underlying event. The CKKW-L matching and merging procedure is applied, with a merging scale of 30 GeV. This setup is hereafter referred to as MG5_aMC@NLO CKKW-L. The renormalisation and factorisation scales are set to the MadGRAPH5_aMC@NLO default values, based on a clustering of the event. The NNPDF3.0nlo PDF set is used, with $\alpha_S = 0.118$. The A14 tune [18] of Pythia 8 is used, and the EvtGen v1.2.0 program [19] is used for properties of the bottom and charm hadron decays. MadGRAPH5_aMC@NLO performs a 5-flavour (5FNS) calculation with massless b - and c -quarks in the matrix element, and massive quarks in the Pythia 8 shower. Samples produced with the described setup, sometimes referred as MG5_aMC+Py8 CKKW-L B in agreement with the definition of [6], corresponds to the main MG5_aMC+Py8 CKKW-L samples used for the studies reported in this note. A second set of samples, which use instead the NNPDF2.3lo PDF set [20] with $\alpha_S = 0.13$, is also available and referred as MG5_aMC+Py8 CKKW-L A, in agreement with the definition of [6].

2.3 MadGRAPH5_aMC@NLO using FxFx

Samples have also been generated using the MadGRAPH5_aMC@NLO program to generate matrix elements for $V + 0, 1$ and 2 partons to NLO accuracy. The showering and subsequent hadronisation has been performed using Pythia 8.210 with the A14 tune, using the NNPDF2.3lo PDF set with $\alpha_S = 0.13$. The different jet multiplicities are merged using the FxFx prescription [21] implemented in the MadGRAPH5_aMC@NLO program (version 2.3.3 is used here).

MG5_aMC@NLO performs a 5FNS calculation with massless b - and c -quarks in the matrix element, and massive quarks in the Pythia8 shower. The PDF input set in use for event generation is NNPDF2.3nlo PDF set with $\alpha_S = 0.119$ and the samples have been generated with additional weights for the PDF replicas as well as scale variations of the renormalisation and the factorisation scales [22].

The impact of various merging scales (μ_Q) has been studied, analysing three different values: 20 GeV (down variation), 25 GeV (nominal value) and 50 GeV (up variation). At the event-generation level, the minimum jet transverse momentum, p_T^j is required to be at least 8 GeV with no restriction on the absolute value of the jet pseudorapidity (jet $|\eta|$). The samples have been generated using LHAPDF-6.1.5 [23] and FastJet-3.1.0 [24].

2.4 POWHEG MINLO

Predictions from POWHEG MINLO [25–27] interfaced to PYTHIA 8.210 [17] with the AZNLO tune [28] were obtained to produce $V + \text{jets}$ events, and will be referred to in the remainder of the note as POWHEG MINLO+PYTHIA8. The EvtGen v1.2.0 program [19] is used as an afterburner to better simulate the decays of bottom and charm hadrons. The PHOTOS++ 3.61 program [29, 30] is interfaced to PYTHIA 8 to accurately describe the QED final state radiation, and the emission of lepton pairs is activated in this setup. The PDF set used in POWHEG is CT14nnlo [31] whereas the PDF set used in the parton shower is the CTEQ6 L1 [32] leading order set.

MINLO is an improvement of the CKKW matching procedure to reach NLO accuracy for boson production in association one jet, and smoothly merges to NLO single boson production at low boson p_T , using no external merging scale. NLO contributions from the CKKW Sudakov form factors are subtracted to the virtual corrections provided by POWHEGBox. Thanks to the optimal scale and to the Sudakov form factors, the POWHEG MINLO samples do not need any Born suppression cut, and their cross-sections are well-behaved at small boson p_T , contrary to fixed order predictions that diverge in this kinematic phase space.

2.5 ALPGEN

Predictions for $V + \text{jets}$ production at 13 TeV have also been produced using the LO matrix-element generator ALPGEN v2.14 [33] interfaced to PYTHIA v6.426 [34] to model the parton shower, hereafter referred to as ALPGEN+PYTHIA6. The parton-shower tune Perugia2011C [35] and the CTEQ6L1 PDF set [32] have been used. Up to five additional partons are modelled by the matrix elements, merged with the MLM prescription [36] with a matching scale of 20 GeV. Different V plus N -partons multiplicities, with N ranging from 0 to 5, are generated in separate samples and, as no distinction is made about the flavour of the parton used in the calculation, the samples are sometimes referred to as “inclusive”.

The predictions follow a four-flavour-number-scheme (4FNS) for the production of heavy-flavour jets. Explicit matrix element evaluation, with massive treatment of the heavy-flavour quarks, is also possible for the following processes (dubbed $V + \text{hf}$):

$$Z + b\bar{b}, Z + c\bar{c}, W + c, W + c\bar{c}, W + b\bar{b}, \quad (1)$$

and, again, a total of up to five partons (generic plus heavy-flavour) can be considered in the calculation.

Each of the $V + \text{hf}$ sample has been produced separately, with up to five partons, and a heavy-flavour overlap removal (HFOR) procedure is used combine the “inclusive” and $V + \text{hf}$ samples. The HFOR is needed to avoid double counting of phase space between different samples because heavy-flavour quarks are both produced by the gluon splitting in the parton shower and by the explicit matrix-element calculations. The HFOR criteria require the removal of an event if it contains two heavy-flavour quarks arising from the PS with $\Delta R(q, \bar{q}) > 0.4$ as well as events which contain two heavy-flavour quarks arising from ME and with $\Delta R(q, \bar{q}) < 0.4$. The pairing of quarks is performed only between same flavours (b or c) and between particle/anti-particle pairs. Quarks produced by multi-parton interactions or originating directly from PDFs are not included in the HFOR procedure. The HFOR procedure reject from about 5% to 30% of the generated events, depending if the analysed sample has lower or larger parton multiplicity or on the considered parton flavour.

The HFOR procedure is applied analysing the HepMC record of each event during a pre-processing stage, before the evaluation of any other kinematic requirement. The predicted cross section of each separate sample is then scaled down to take into account the removed fraction of events. The total cross section predicted by the combination of all the samples is in agreement within 1% with the prediction of the “inclusive” only samples. Several kinematic variables are affected by the combination of the “inclusive” and $V + \text{hf}$ samples, however a net effect immediately visible is an enhancement of the differential cross section for jet multiplicities ≥ 1 .

2.6 Higher-order cross sections

In this note, all absolute differential cross section have been normalised to an NNLO-accurate cross section calculated with FEWZ [37]. The NNLO cross section at 8 and 13 TeV are given in Table 1.

Table 1: NNLO cross section at 8 and 13 TeV. The Z/γ^* cross sections are quoted for the dilepton mass window $66 \text{ GeV} < m_{\ell\ell} < 116 \text{ GeV}$.

Process	$\sigma(8 \text{ TeV}) [\text{pb}]$	$\sigma(13 \text{ TeV}) [\text{pb}]$
$W^\pm$ production	12100 ± 600	20100 ± 1000
Z/γ^* production	1122 ± 56	1906 ± 95

3 Rivet routines for comparison

Particle-level comparisons of different generator setups are made using Rivet v2.5.3 [38] using the following analyses:

- ATLAS 13 TeV analysis of Z+ jets production using an integrated luminosity of 3.16 fb^{-1} [39] (not yet published in Rivet). A pair of opposite-charge dressed leptons, with $p_T > 25 \text{ GeV}$ and $|\eta| < 2.5$ are required, with dilepton mass between 66 and 116 GeV. Jets are constructed using the anti- k_t algorithm [13] using a jet-radius parameter of 0.4, and required to have $p_T > 30 \text{ GeV}$, $|y| < 2.5$ and not to overlap with the selected leptons ($\Delta R(j, \ell) > 0.4$).
- ATLAS 8 TeV measurement of the splitting scales (ATLAS_2017_I1589844) occurring in the k_t algorithm [40, 41] using 20.2 fb^{-1} [42]. In the electron channel, exactly two dressed electrons are selected with $p_T > 25 \text{ GeV}$ and $|\eta| < 2.47$, excluding the transition region between the ATLAS barrel and end cap electromagnetic calorimeters ($1.37 < |\eta| < 1.52$). The dilepton invariant mass is required to be within $71 \text{ GeV} < m_{\ell\ell} < 111 \text{ GeV}$. All charged final-state particles with $p_T > 400 \text{ MeV}$ and $|\eta| < 2.5$, excluding the tracks of the two selected electrons, are then passed through the k_t algorithm and its splitting scales are measured for jet-radius parameters at $R = 0.4$ as well as $R = 1.0$.
- Analysis of $V + b(b)$ events in different phase space regions: an *inclusive* region, aimed at the differential measurement of W and Z production in association with heavy-flavour jets, a *high- p_T selection*, aimed at understanding this type of events when the two b -jets are almost collinear and have a very high boost as well as an *intermediate* region, aimed at describing the main irreducible background of the $VH(H \rightarrow b\bar{b})$ process. Jets originating from b -quarks are identified by the presence of a weakly decaying b -hadron with $p_T > 5 \text{ GeV}$ and decay products within the area of the jet (ghost-matching). This procedure mimics a b -tagging algorithm with efficiency of 100%, therefore such jet selection is often referred as b -tagging in the following. Further details about the mentioned analyses will be given in the relevant comparison sections.
- Analysis of electroweak Wjj production, based on the ATLAS 8 TeV measurement [43]. One lepton with $p_T > 25 \text{ GeV}$ and $|\eta| < 2.5$ is required, together with a missing energy larger than 20 GeV. The m_T is required to be larger than 40 GeV and the two associated jets to have a transverse momentum of 80 and 60 GeV, respectively for the leading and subleading one, and to be within 4.4 in rapidity. Additional topological cuts are applied to the jet pair, requiring their invariant mass (m_{jj}) to be larger than 500 GeV, and $\Delta y_{jj} > 2$ and $\Delta R_{j\ell} > 0.3$.
- ATLAS 8 TeV measurement of electroweak dijet production in association with a Z boson (Zjj) [44]. The corresponding routine (ATLAS_2014_I1279489) has been modified and is used to compare the 13 TeV samples (without data). The selection remains the same, but the binning has been adjusted to suit the reach of the 13 TeV predictions.
- Analysis of $W + \text{jets}$ final states in the $W \rightarrow e\nu$ channel. This routine was initially designed for a centre-of-mass energy of 8 TeV, but is used in the context of this note to study the 13 TeV predictions. An electron is selected and needs to be within $|\eta| < 2.47$ and have $p_T > 25 \text{ GeV}$. A symmetric cut on the missing energy at 25 GeV is applied, and the transverse mass is required to be above 40 GeV. Only jets with $p_T > 30 \text{ GeV}$ and $|y| < 4.4$ are considered in the analysis.

4 W and Z boson plus jets

The most commonly used generators by ATLAS are compared using realistic V + jets selections. Generator comparisons for V + heavy-flavour jets are made separately from those using inclusive V + jets selections.

4.1 Comparison between different generators

4.1.1 $Z(\rightarrow \ell\ell)$ + jets

Figures 1–4 show comparisons of SHERPA 2.2 (red), POWHEG MINLO+PYTHIA8 (blue) and MG5_aMC@NLO+Py8 using FxFx (green) against data (black) from the 13 TeV ATLAS Z +jets measurement [39]. The measurement uncertainty is indicated by a grey band in the ratio panels, while generator uncertainties have been estimated using SHERPA 2.2 and are indicated by the orange band in the ratio panels. This theory uncertainty band is constructed from the statistical uncertainty of the SHERPA predictions, the PDF uncertainties from the NNPDF3.0nnlo replicas estimated using LHAPDF [45] along with α_s variations of ± 0.002 around the nominal value of 0.118 as well as an envelope formed from 7-point scale variations of the renormalisation and factorisation scales. The various uncertainty components are added in quadrature to give the total uncertainty shown.

Figure 1 shows the inclusive and exclusive jet multiplicity distributions. The MEPS@NLO setup using SHERPA 2.2 is NLO-accurate for up to two extra emissions and LO-accurate for up to four extra emissions. At very high jet multiplicities, SHERPA 2.2 starts to diverge from the data, indicating too much activity in the parton shower. The MG5_aMC@NLO setup already begins to mismodel the data sooner, as it lacks additional multilegs beyond the third emission, making it less attractive for many new-physics searches interested in a good modelling of high multijet configurations, although the mismodelling is likely covered by the scale uncertainties which are on the order of 30 to 40 %.

Similar comparisons of the differential cross sections as a function of the scalar jet- p_T sum, H_T , and the leading jet rapidity are shown in Figure 2, as a function of the exclusive and the inclusive leading jet p_T in Figure 3 and as function of the azimuthal difference between the leading two jets, $\Delta\phi_{jj}$, and the dijet invariant mass in Figure 4.

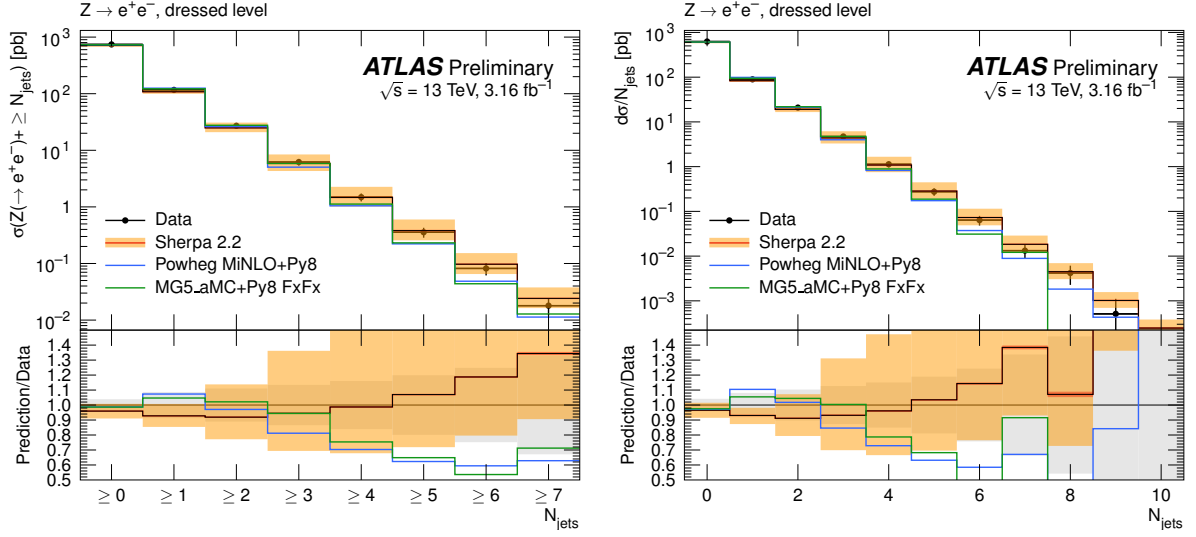


Figure 1: Predictions for the differential cross sections as a function of the inclusive (left) and the exclusive (right) jet multiplicity from SHERPA 2.2 (red), POWHEG MiNLO+PYTHIA8 (blue) and MG5_aMC@NLO+Py8 using FxFx (green). The predictions are compared to data (black) from a recent ATLAS measurement of Z+ jets production [39]. The grey band indicates the size of the total measurement uncertainty, while the orange band includes PDF and scale uncertainties estimated with SHERPA 2.2. The size of the statistical uncertainty components are indicated by the size of the error bars.

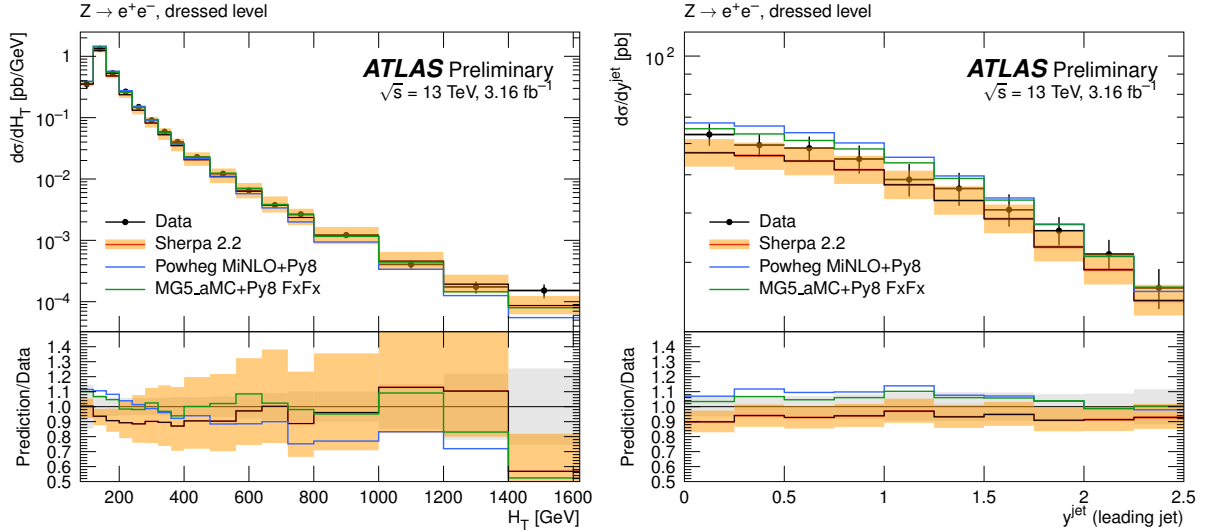


Figure 2: Predictions for the differential cross sections as a function of H_T (left) and the leading jet rapidity (right) from SHERPA 2.2 (red), POWHEG MiNLO+PYTHIA8 (blue) and MG5_aMC@NLO+Py8 using FxFx (green). The predictions are compared to data (black) from a recent ATLAS measurement of Z+ jets production [39]. The grey band indicates the size of the total measurement uncertainty, while the orange band includes PDF and scale uncertainties estimated with SHERPA 2.2. The size of the statistical uncertainty components are indicated by the size of the error bars.

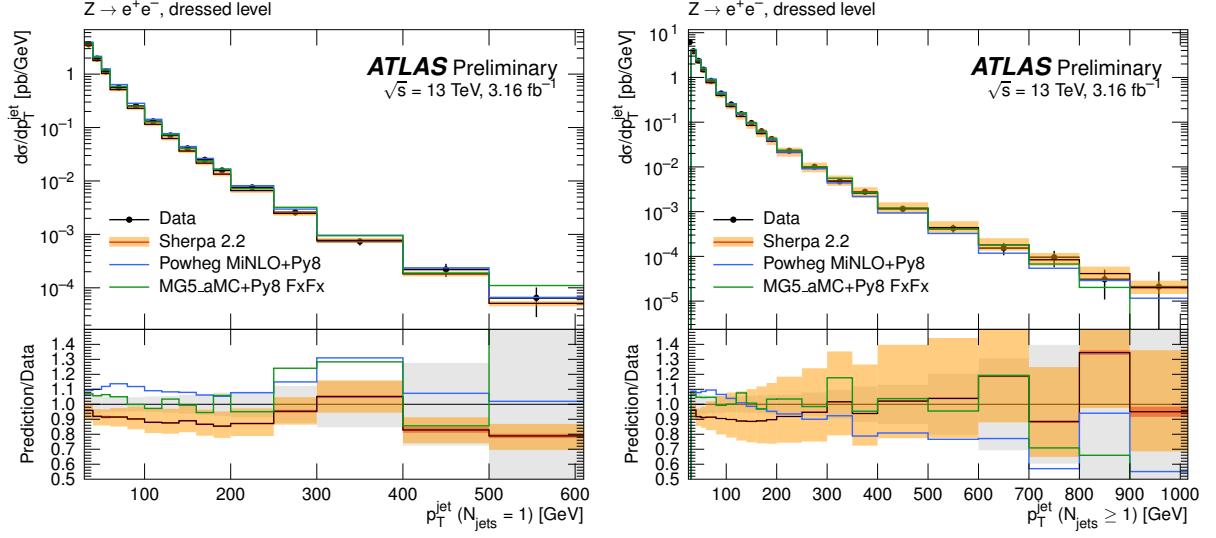


Figure 3: Predictions for the differential cross sections as a function of the exclusive (left) and (inclusive) leading jet p_T from SHERPA 2.2 (red), POWHEG MiNLO+PYTHIA8 (blue) and MG5_aMC@NLO+Py8 using FxFx (green). The predictions are compared to data (black) from a recent ATLAS measurement of Z+ jets production [39]. The grey band indicates the size of the total measurement uncertainty, while the orange band includes PDF and scale uncertainties estimated with SHERPA 2.2. The size of the statistical uncertainty components are indicated by the size of the error bars.

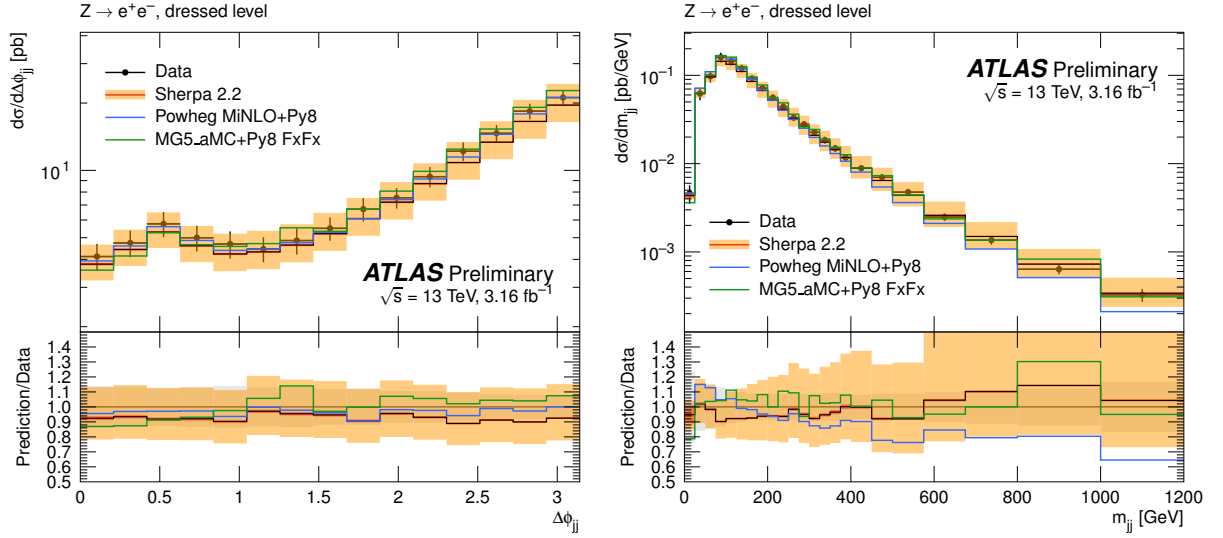


Figure 4: Predictions for the differential cross sections as a function of the $\Delta\phi_{jj}$ (left) and the dijet invariant mass (right) from SHERPA 2.2 (red), POWHEG MiNLO+PYTHIA8 (blue) and MG5_aMC@NLO+Py8 using FxFx (green). The predictions are compared to data (black) from a recent ATLAS measurement of Z+ jets production [39]. The grey band indicates the size of the total measurement uncertainty, while the orange band includes PDF and scale uncertainties estimated with SHERPA 2.2. The size of the statistical uncertainty components are indicated by the size of the error bars.

A comparison to 8 TeV data is also made using the ATLAS measurement of the splitting scales occurring in the k_t algorithm [42]. Figures 5 and 6 are showing the first-order splitting scale, $\sqrt{d_1}$, and the third-order splitting scale, $\sqrt{d_3}$, using jet-radius parameters of 0.4 and 1.0. Both setups seem to struggle with the transition region between the perturbative and the non-perturbative regimes. In general, SHERPA 2.2 does slightly better than the MG5_aMC@NLO+PYTHIA8 with FxFx setup for the high-energy part of the splitting-scale spectrum, whereas the soft region, which is mainly sensitive to the parton shower models, seems to be generally better described by PYTHIA8 than the SHERPA 2.2 parton shower. The measurement uncertainty in the transition and soft region is small enough to allow for the data to be used to better constrain the modelling in the future.

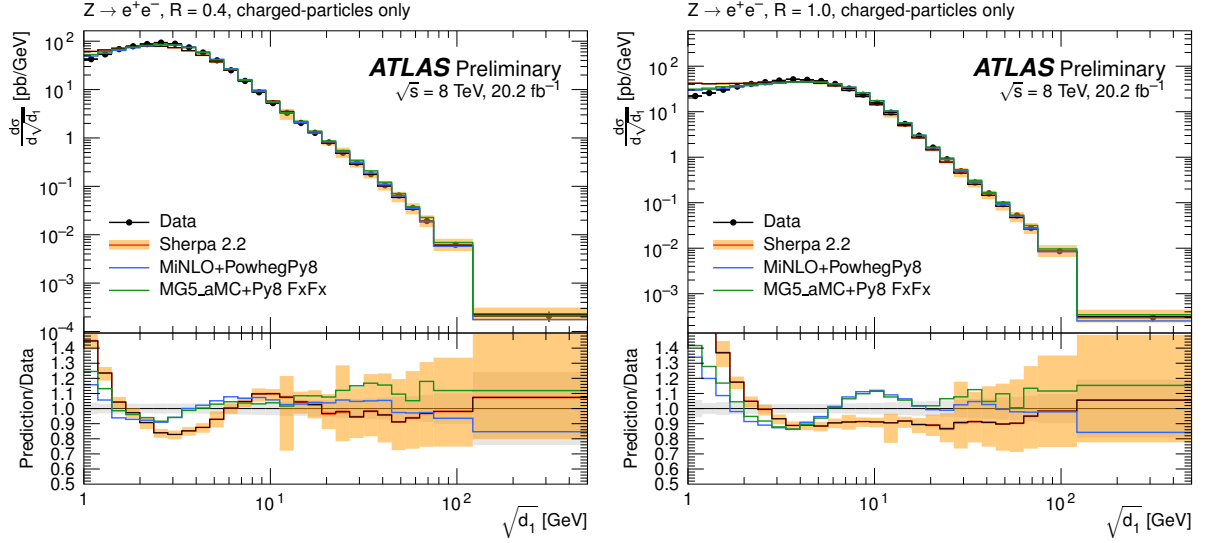


Figure 5: Predictions for the differential cross sections as a function of first-order splitting scale, $\sqrt{d_1}$, in the k_t algorithm using a jet-radius parameter of $R = 0.4$ (left) and $R = 1.0$ (right) from SHERPA 2.2 (red), POWHEG MiNLO+PYTHIA8 (blue) and MG5_aMC@NLO+Py8 using FxFx (green). The predictions are compared to data (black) from a recent ATLAS measurement of the k_t splitting scales [42]. The grey band indicates the size of the total measurement uncertainty, while the orange band includes PDF and scale uncertainties estimated with SHERPA 2.2. The size of the statistical uncertainty components are indicated by the size of the error bars.

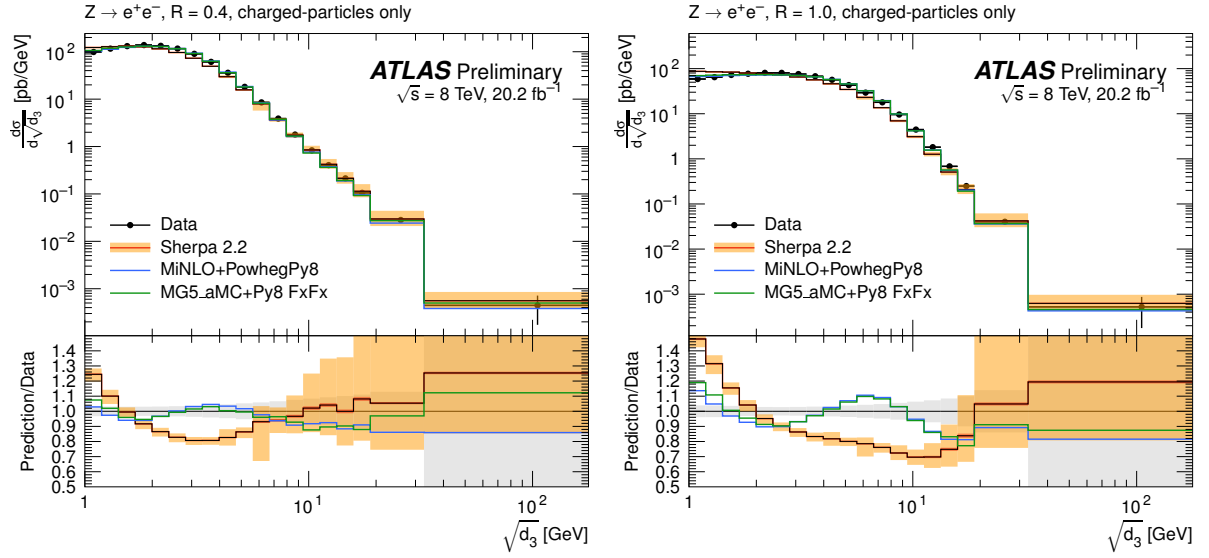


Figure 6: Predictions for the differential cross sections as a function of third-order splitting scale, $\sqrt{d_3}$, in the k_t algorithm using a jet-radius parameter of $R = 0.4$ (left) and $R = 1.0$ (right) from SHERPA 2.2 (red), POWHEG MiNLO+PYTHIA8 (blue) and MG5_aMC@NLO+Py8 using FxFx (green). The predictions are compared to data (black) from a recent ATLAS measurement of the k_t splitting scales [42]. The grey band indicates the size of the total measurement uncertainty, while the orange band includes PDF and scale uncertainties estimated with SHERPA 2.2. The size of the statistical uncertainty components are indicated by the size of the error bars.

4.1.2 $W(\rightarrow \ell\nu)+\text{jets}$

Figures 7–10 show generator comparisons of $W+\text{jets}$ predictions from SHERPA 2.2 and MG5_aMC@NLO+Py8 using CKKW-L (blue).

Generator uncertainties have been estimated using SHERPA 2.2 and are indicated by the orange band in the ratio panels. The band is constructed from the statistical uncertainty of the SHERPA predictions, the PDF uncertainties from the NNPDF3.0nnlo replicas estimated using LHAPDF along with α_S variations of ± 0.002 around the nominal value of 0.118 as well as an envelope formed from 7-point scale variations of the renormalisation and factorisation scales. The various uncertainty components are added in quadrature to give the total uncertainty shown.

In Figure 7, it can be seen that the uncertainty for the MEPS@NLO setup is at the few-percent level for the first couple of jet bins that have NLO accuracy, then rises to about 20 % for the LO-accurate jet multiplicities and finally becomes very large for high jet multiplicities that are dominated by the parton shower.

Similar comparisons of the differential cross sections as a function of the leading and subleading jet transverse momentum are shown in Figure 8, as a function of the lepton transverse momentum and the missing transverse energy, MET, in Figure 9 as well as the differential cross sections as a function of the boson transverse momentum for $N_{\text{jets}} \geq 0$ and $N_{\text{jets}} \geq 1$ in Figure 10. The unphysical spikes in the distributions shown in Figure 9 are due to very large event weights that can occur if the differential cross section of a phase-space point in the Monte Carlo integration exceeds the maximum weight estimated during the phase-space optimisation.

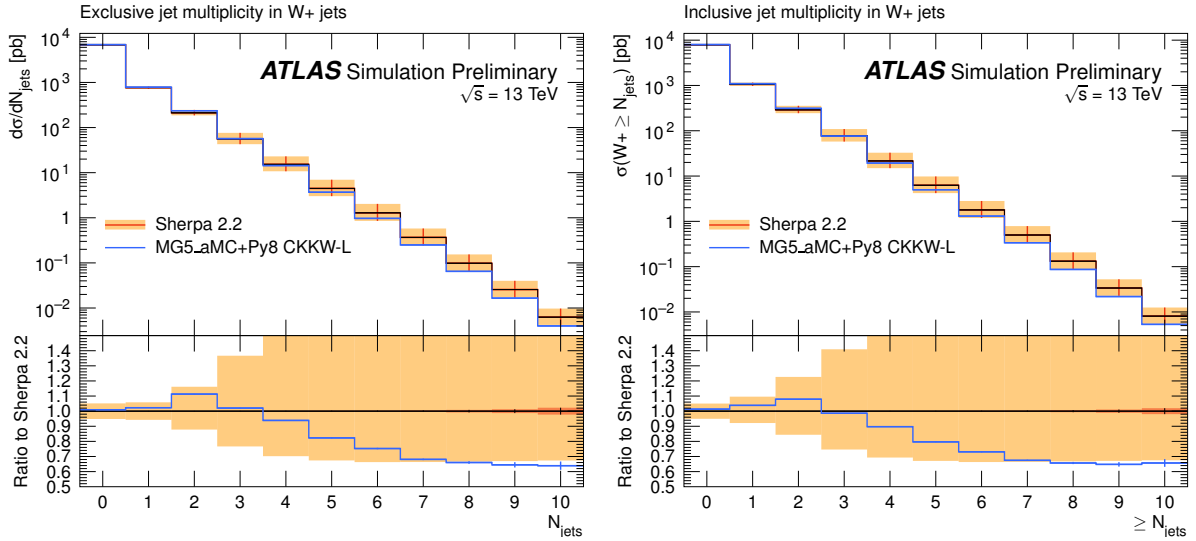


Figure 7: Predictions for the cross sections as a function of the exclusive (left) and inclusive (right) jet multiplicities from SHERPA 2.2 (red) and MG5_aMC@NLO+Py8 using CKKW-L (blue). The orange band includes PDF and scale uncertainties estimated with SHERPA 2.2, while the size of the statistical uncertainty components are indicated by the size of the error bars.

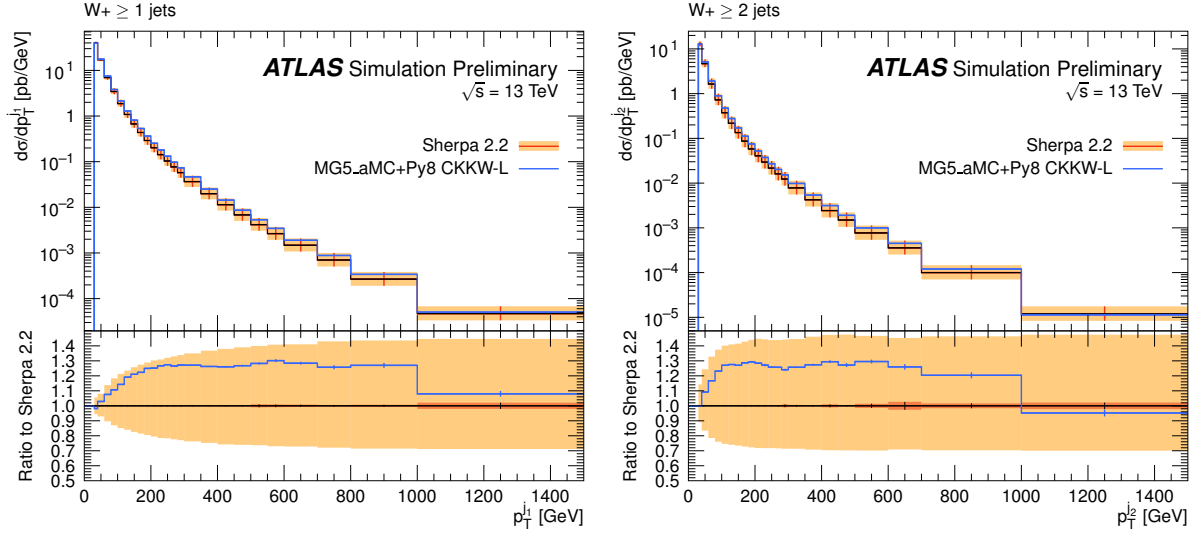


Figure 8: Predictions for the differential cross sections as a function of the leading (left) and subleading (right) jet transverse momentum from SHERPA 2.2 (red) and MG5_aMC@NLO+Py8 using CKKW-L (blue). The orange band includes PDF and scale uncertainties estimated with SHERPA 2.2, while the size of the statistical uncertainty components are indicated by the size of the error bars.

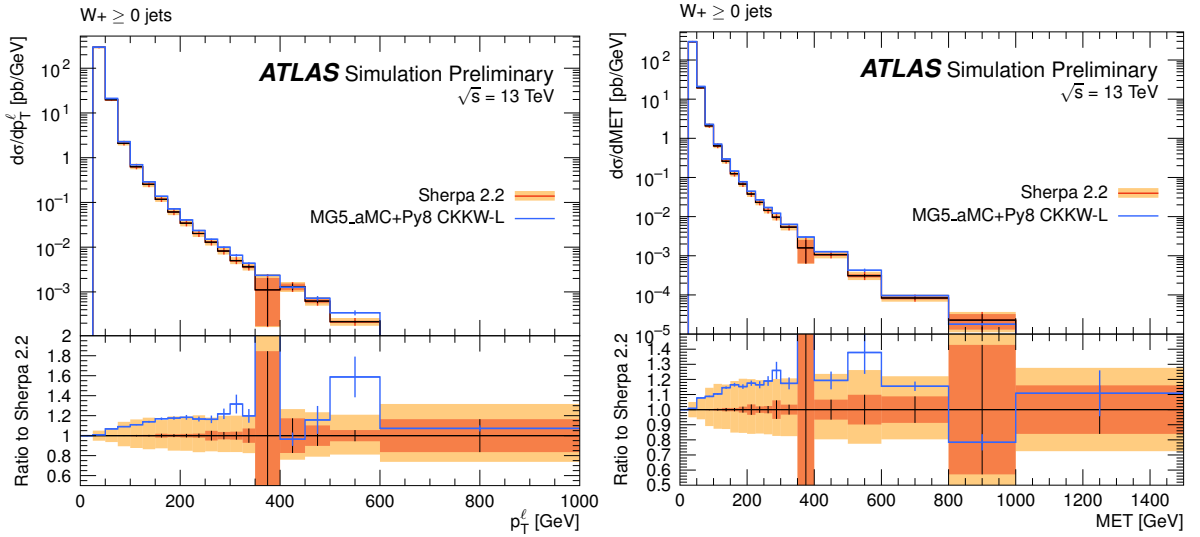


Figure 9: Predictions for the differential cross sections as a function of the lepton transverse momentum (left) and the missing transverse energy (right) from SHERPA 2.2 (red) and MG5_aMC@NLO+Py8 using CKKW-L (blue). The orange band includes PDF and scale uncertainties estimated with SHERPA 2.2, while the size of the statistical uncertainty components are indicated by the size of the error bars.

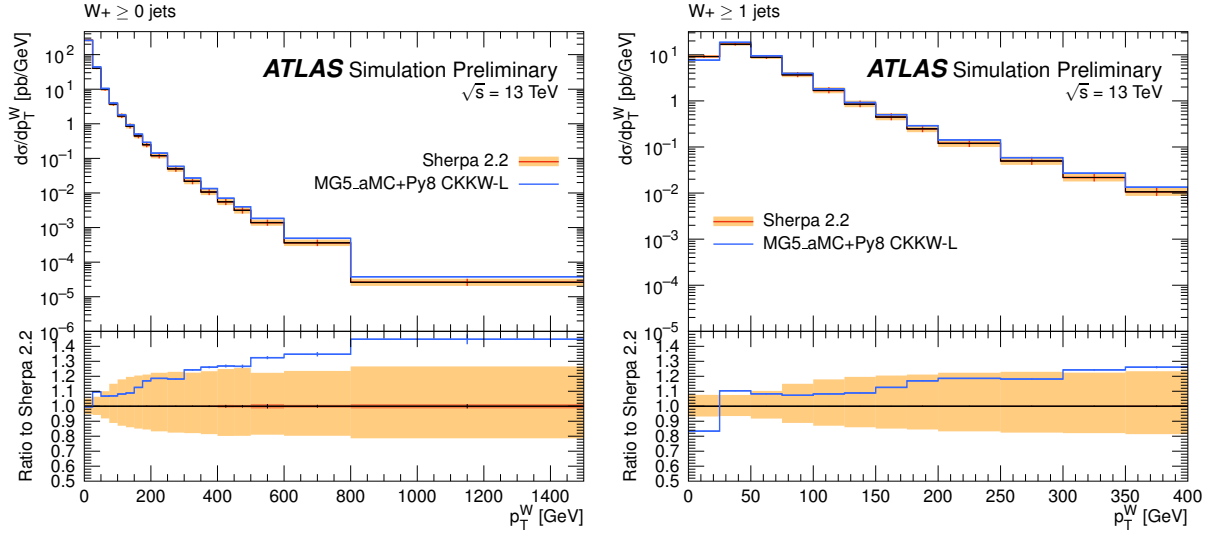


Figure 10: Predictions for the differential cross sections as a function of the boson transverse momentum for $N_{\text{jets}} \geq 0$ (left) and $N_{\text{jets}} \geq 1$ (right) from SHERPA 2.2 (red) and MG5_aMC@NLO+Py8 using CKKW-L (blue). The orange band includes PDF and scale uncertainties estimated with SHERPA 2.2, while the size of the statistical uncertainty components are indicated by the size of the error bars.

4.2 Systematic uncertainties

Figure 11 shows generator comparisons of SHERPA 2.2 (red) with MG5_aMC@NLO+Py8 using FxFx (blue) at a centre-of-mass energy of 13 TeV. Generator uncertainties that have been estimated using SHERPA 2.2 and are indicated by the orange band in the ratio panels. This theory uncertainty band is constructed from the statistical uncertainty of the SHERPA 2.2 predictions, the PDF uncertainties from the NNPDF3.0nnlo replicas estimated using LHAPDF along with α_S variations of ± 0.002 around the nominal value of 0.118 as well as an envelope formed from 7-point scale variations of the renormalisation and factorisation scales. The various uncertainty components are added in quadrature to give the total uncertainty shown. In addition, variations of the FxFx matching scale, μ_Q , (shown in solid and dashed green) have been studied for the MG5_aMC@NLO+Py8 setup. The nominal prediction uses a matching scale of 25 GeV and variations using matching scales of 20 GeV and 50 GeV are shown in solid and dashed green lines, respectively. The statistical uncertainty component is indicated by the error bars.

Figures 12 and 13 show similar comparisons for the first- and third-order splitting scale distribution occurring in the k_t algorithm. It can be seen that neither the PDF and scale variations estimated using SHERPA 2.2 nor the matching-scale variations estimated for the FxFx setup can cover the differences between the two predictions seen in the transition region between the soft and the perturbative regimes.

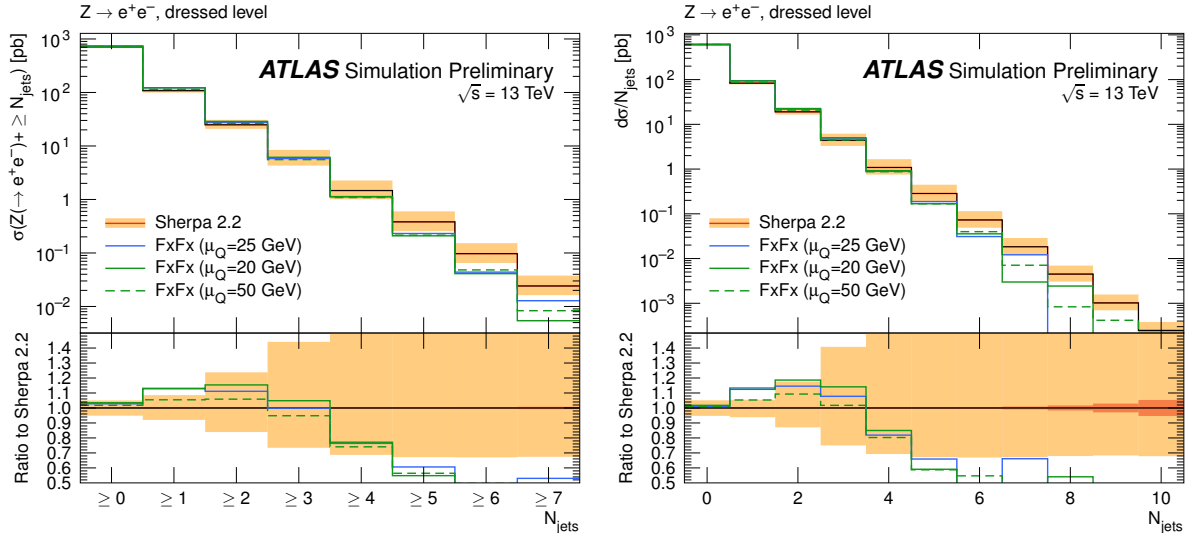


Figure 11: Predictions for the differential cross sections as a function of the inclusive (left) and the exclusive (right) jet multiplicity from SHERPA 2.2 (red) and MG5_aMC@NLO using FxFx using matching scales of 20 GeV (blue), 25 GeV (solid green) and 50 GeV (dashed green). The yellow band includes statistical, PDF and scale uncertainties estimated with SHERPA 2.2, while the error bars indicate the size of the statistical component.

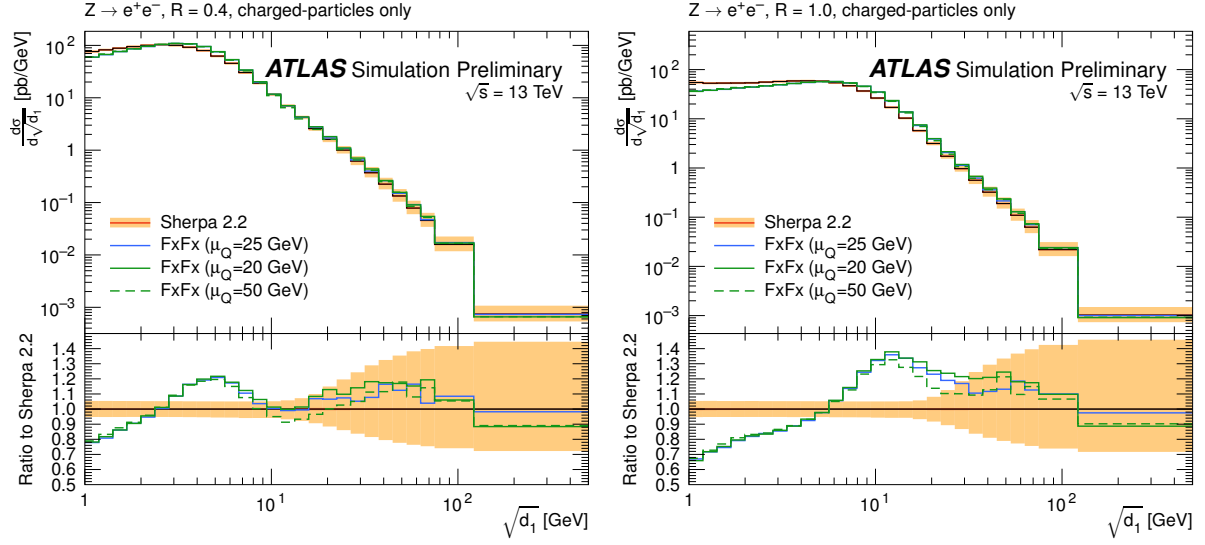


Figure 12: Predictions for the differential cross sections as a function of first-order splitting scale, $\sqrt{d_1}$, in the k_t algorithm using a jet-radius parameter of $R = 0.4$ (left) and $R = 1.0$ (right) from SHERPA 2.2 (red) and MG5_aMC@NLO using FxFx using matching scales of 20 GeV (blue), 25 GeV (solid green) and 50 GeV (dashed green). The yellow band includes statistical, PDF and scale uncertainties estimated with SHERPA 2.2, while the error bars indicate the size of the statistical component.

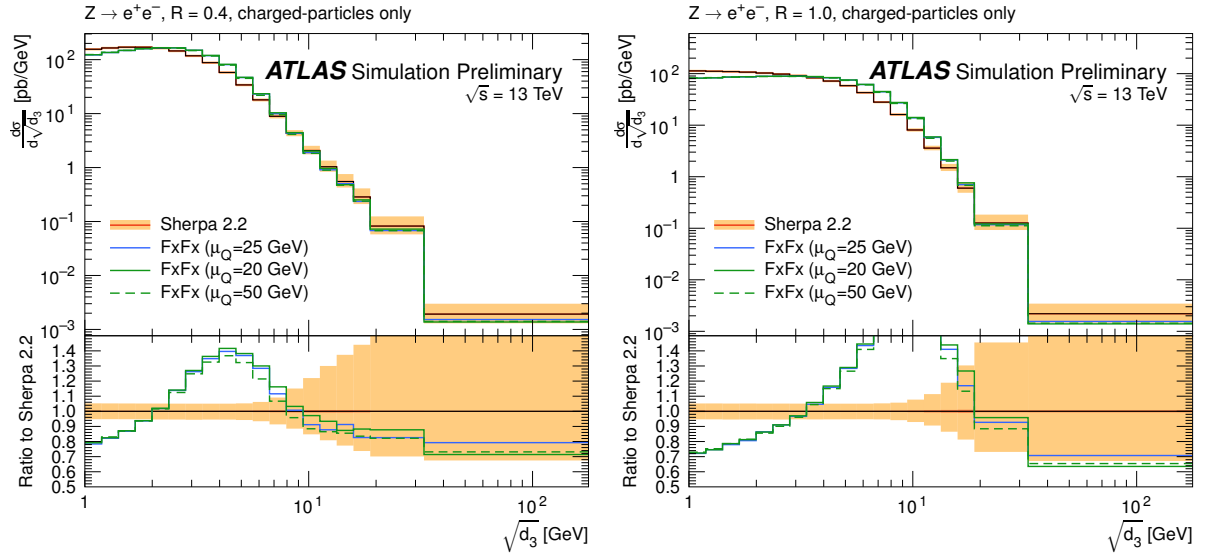


Figure 13: Predictions for the differential cross sections as a function of third-order splitting scale, $\sqrt{d_3}$, in the k_t algorithm using a jet-radius parameter of $R = 0.4$ (left) and $R = 1.0$ (right) from SHERPA 2.2 (red) and MG5_aMC@NLO using FxFx using matching scales of 20 GeV (blue), 25 GeV (solid green) and 50 GeV (dashed green). The yellow band includes statistical, PDF and scale uncertainties estimated with SHERPA 2.2, while the error bars indicate the size of the statistical component.

5 W and Z boson plus heavy-flavour jets

The $V + b(b)$ production processes are an extremely interesting benchmark for MC simulations because the predicted cross-section rates and distributions give an insight on both the structure of the proton and on the gluon splitting into a pair of b -quarks. The $V + b(b)$ processes are also one of the largest and most difficult to model backgrounds for searches at hadron colliders. This is reflected particularly in the case where the $p_T(V)$ is very large, and in the case of the search for the SM $VH(H \rightarrow b\bar{b})$ process, where they constitute the main irreducible background, and the uncertainty in their predictions largely affects the result. These aspects will be treated in the following, starting by looking at important kinematic variables for the $V + b(b)$ process measurements, then studying the modelling in case of a large boost of the di- b -jet system, and subsequently in the context of a SM $VH(H \rightarrow b\bar{b})$ analysis, where it is often hard to identify in data a control region pure enough in $V + b(b)$ production, making it necessary to carefully understand the available predictions.

In this section the V +jets generators described in Section 2 are compared, at a centre-of-mass energy of 13 TeV, using custom Rivet routines aimed at selecting final states with b -jets, specifically:

- b. $W(\rightarrow \ell\nu) + b(b)$
- a. $Z(\rightarrow \ell\ell) + b(b)$
- c. $Z(\rightarrow \nu\nu) + b(b)$.

For all the three mentioned processes we examined the case of b -jets reconstructed using the anti- k_t algorithm, with a radius parameter R of 0.4. For the second process, it is examined also the case of one single anti- k_t jet of $R = 1$ (large-radius jet) having two associated b -quarks, and the p_T of the large-radius jet is taken to be larger than 250 GeV.

Predictions for $V + b(b)$ processes in the 5FNS are available using the SHERPA 2.2 and MG5_aMC+Py8 CKKW-L samples, while prediction in the 4FNS are available using the Alpgen samples.

5.1 $W(\rightarrow \ell\nu) + b(b)$ with anti- k_t $R=0.4$ b -jets

The event selection used to identify the $W(\rightarrow e\nu) + b(b)$ process, in the case where two b -quarks form separate b -jets is described in the following. The electron, is selected only if its p_T is larger than 25 GeV, and $|\eta| < 2.5$. A requirement of $p_T > 25$ GeV is also applied to the neutrino in the event, and subsequently the W boson transverse mass is required to be larger than 60 GeV. Events with exactly one or two b -tagged jets (as described in Section 3) with $p_T > 20$ GeV and $|\eta| < 2.5$, and no additional jets with $p_T > 20$ GeV and $|\eta| < 4.5$ are selected. The events are also discarded if the ΔR between any jet and the electron is below 0.4.

Table 5.1 reports the predicted cross sections rates while, Figures 14–16 show some relevant kinematic variables in $W + b(b)$ events, for the different generators considered and for a centre of mass energy of 13 TeV. NLO-accurate predictions from SHERPA 2.2, which uses the 5FNS, are reported in red, together with the systematic uncertainty associated to them as the orange band, and compared to the LO-accurate predictions from MG5_aMC@NLO+Py8 CKKW-L B (in blue, using 5FNS) and ALPGEN+PYTHIA6 (in purple, using 4FNS). The statistical uncertainty is shown as an error bar for each bin. The SHERPA uncertainty band is constructed from the statistical uncertainty of the predictions, the PDF uncertainties from the NNPDF3.0nnlo replicas estimated using LHAPDF along with α_S variations of ± 0.002 around the

nominal value of 0.118 as well as an envelope formed from 7-point scale variations of the renormalisation and factorisation scales. The various uncertainty components are added in quadrature to give the total uncertainty shown in the plots. The ratio to SHERPA is shown as default in the plots.

Generator	$\sigma_{W+1b\text{jet}}$ [pb]	$\sigma_{W+2b\text{jets}}$ [pb]
SHERPA 2.2	36.1 ± 0.1	4.39 ± 0.03
ALPGEN+PYTHIA6	20.08 ± 0.05	3.13 ± 0.01
MG5_aMC@NLO+Py8 CKKW-L	20.36 ± 0.06	3.26 ± 0.01

Table 2: Predicted cross section for the associate production of a W boson and exactly one or two b -jets events for the following MC generators, described in Section 2: SHERPA 2.2 (5FNS) ALPGEN (4FNS), and MG5_aMC@NLO+Py8 (CKKW-L) (5FNS). Uncertainties are due to the MC statistics used in the simulation.

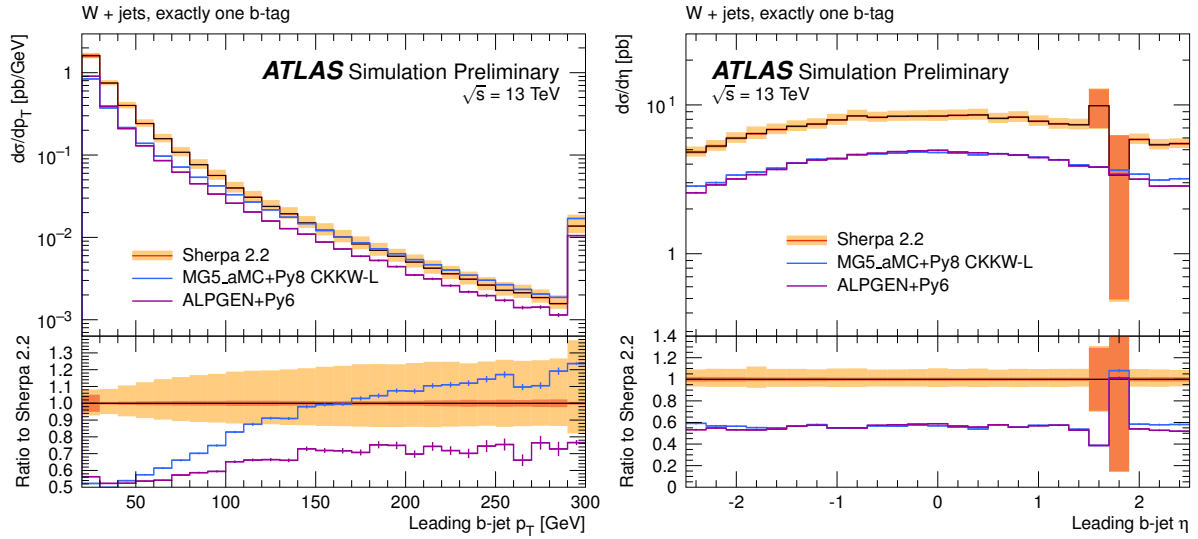


Figure 14: Predictions for the differential cross sections as a function of the b -jet p_T (left) and η (right) in events with exactly one b -tagged jet from SHERPA 2.2 (red), MG5_aMC@NLO+Py8 using CKKW-L (blue) and ALPGEN+PYTHIA6 (purple). The orange band includes PDF and scale uncertainties estimated with SHERPA 2.2, while the size of the statistical uncertainty components are indicated by the size of the error bars.

The cross section from the NLO-accurate SHERPA prediction is systematically higher than that of the two LO-accurate predictions, both in the case of one and two b -jets in the final state. MG5_aMC@NLO+Py8 predicts a substantially harder jet transvers momentum of the vector boson than the other two predictions, and the uncertainties estimated on SHERPA do not cover the slope difference. Very similar shapes are observed for the three predictions for the angular variables in the case of one- b -jet final states. Some effects of large weights for SHERPA, already discussed in Sec 4.1.2 are observed in the three angular distributions for the one- b -jet selection. The predictions for the di- b -jet variables show discrepancies between the shape of MG5_aMC+Py8 CKKW-L and the other two predictions, which have instead a similar behaviour, for the invariant mass of the di- b -jet pair. The ΔR_{bb} distribution, instead, shows a very similar modelling for the two LO predictions, which differs significantly for the NLO prediction from SHERPA.

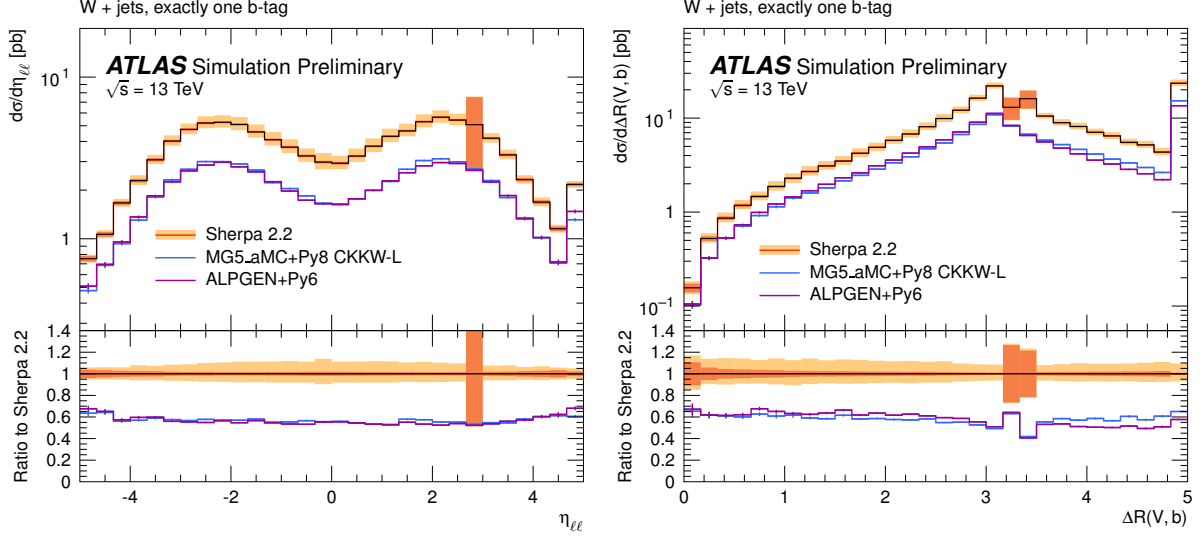


Figure 15: Predictions for the differential cross sections as a function of the pseudorapidity of the W boson (left) and of the angular distance between the lepton-neutrino system and the b -tagged jet, $\Delta R(V, b)$, (right) in events with exactly one b -tagged jet from SHERPA 2.2 (red), MG5_aMC@NLO+Py8 using CKKW-L (blue) and ALPGEN+PYTHIA6 (purple). The orange band includes PDF and scale uncertainties estimated with SHERPA 2.2, while the size of the statistical uncertainty components are indicated by the size of the error bars.

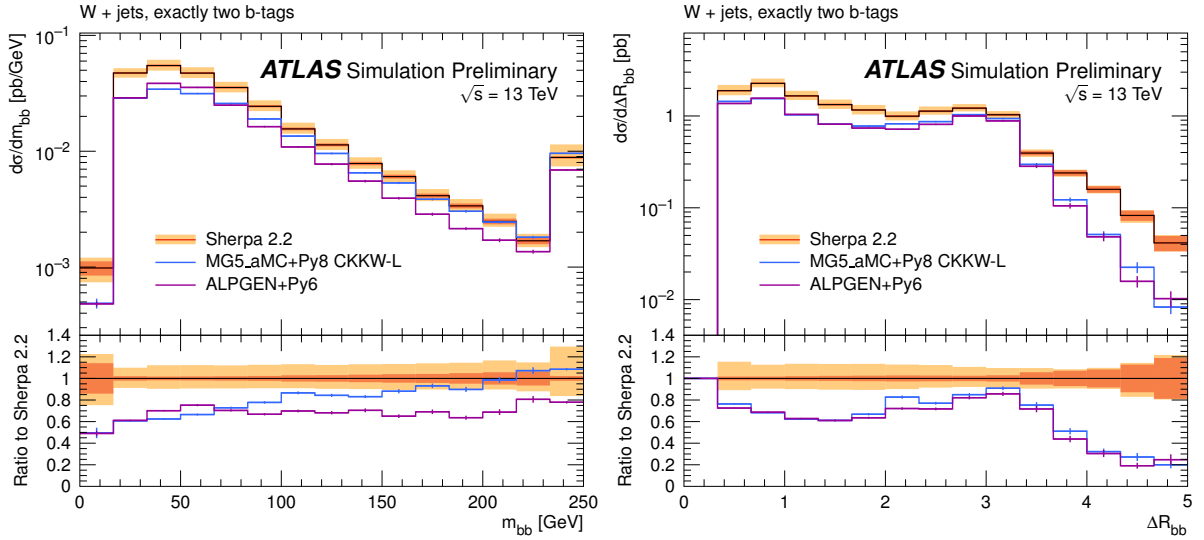


Figure 16: Predictions for the differential cross sections as a function of the invariant mass of two jets, m_{bb} , (left) and the angular separation between them, ΔR_{bb} , (right) in events with exactly two b -tagged jets from SHERPA 2.2 (red), MG5_aMC@NLO+Py8 using CKKW-L (blue) and ALPGEN+PYTHIA6 (purple). The orange band includes PDF and scale uncertainties estimated with SHERPA 2.2, while the size of the statistical uncertainty components are indicated by the size of the error bars.

5.2 $Z(\rightarrow \ell\ell) + b(b)$ with one large- R jet

This selection probes $Z(\rightarrow ee) + b(b)$ events with very energetic hadronic activity. Two electrons with $p_T > 25$ GeV and $|\eta| < 2.5$ are required, and their invariant mass must lie near the Z boson mass pole $71 < m_{\ell\ell} < 111$ GeV. In addition, there must be exactly one anti- k_t $R = 1.0$ ("large- R ") jet built from stable particles with $p_T > 250$ GeV. Anti- k_t $R = 0.2$ jets built from charged stable particles ("track-jets") are used to categorize the flavor content of the large- R jets. Track-jets are ghost-associated to large- R jets, and the large- R jets are categorized by the number of associated track-jets, b -tagged according to the description of Sec 3.

Figure 17 shows the large- R jet mass and transverse momentum spectra for such events with no requirement on the flavor content of the large- R jet. Figure 18 shows predictions for a region of phase space with at least two b -quark labelled track-jets associated to the large- R jet: the angular distance between the two b -flavored track-jets and the large- R jet p_T . NLO predictions from SHERPA 2.2, which uses the 5FNS, are reported in red, together with the PDF and scale uncertainties as indicated by the orange band and the statistical uncertainty indicated by the red bars in each bin. The LO predictions from MG5_aMC@NLO+Py8 using CKKW-L, which also use the 5FNS, are shown in blue and predictions using the 4FNS from ALPGEN+PYTHIA6 are shown in purple. The SHERPA uncertainty band is constructed from the statistical uncertainty of the predictions, the PDF uncertainties from the NNPDF3.0nnlo replicas estimated using LHAPDF along with α_S variations of ± 0.002 around the nominal value of 0.118 as well as an envelope formed from 7-point scale variations of the renormalisation and factorisation scales. The various uncertainty components are added in quadrature to give the total uncertainty shown in the plots. The ratio to SHERPA is shown in the figures.

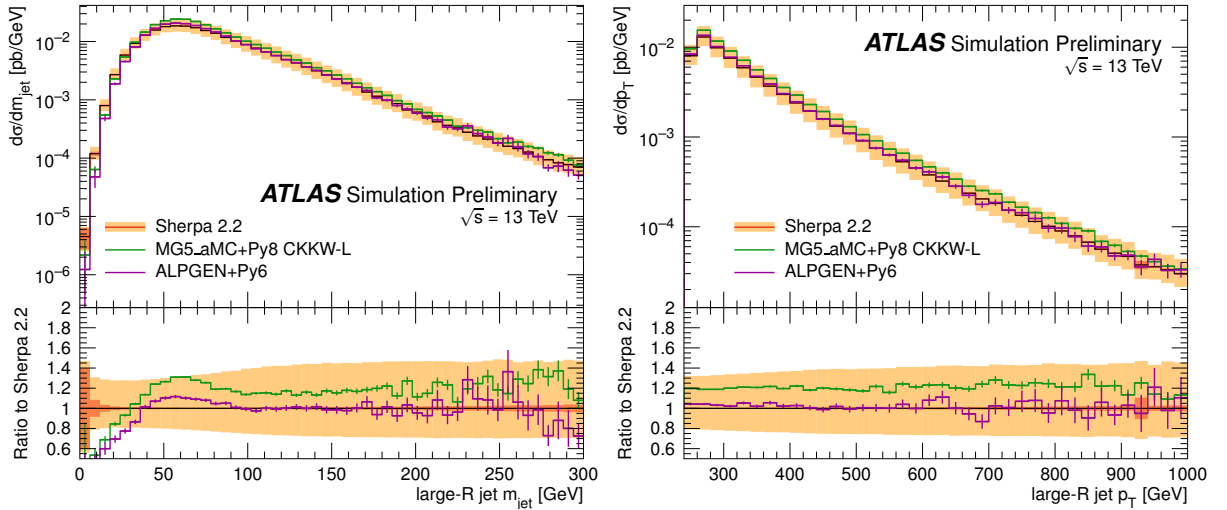


Figure 17: Predictions for the differential cross sections as a function of the invariant mass (left) and the transverse momentum of the large- R jet in events with no requirement on the number of b -tagged jets from SHERPA 2.2 (red), MG5_aMC@NLO+Py8 using CKKW-L (blue) and ALPGEN+PYTHIA6 (purple). The orange band includes PDF and scale uncertainties estimated with SHERPA 2.2, while the size of the statistical uncertainty components are indicated by the size of the error bars.

The predicted cross section in this specific phase-space is compatible for ALPGEN+PYTHIA6 and SHERPA in the non- b -tagged region, but MG5_aMC+Py8 CKKW-L shows a 20% discrepancy in normalisation,

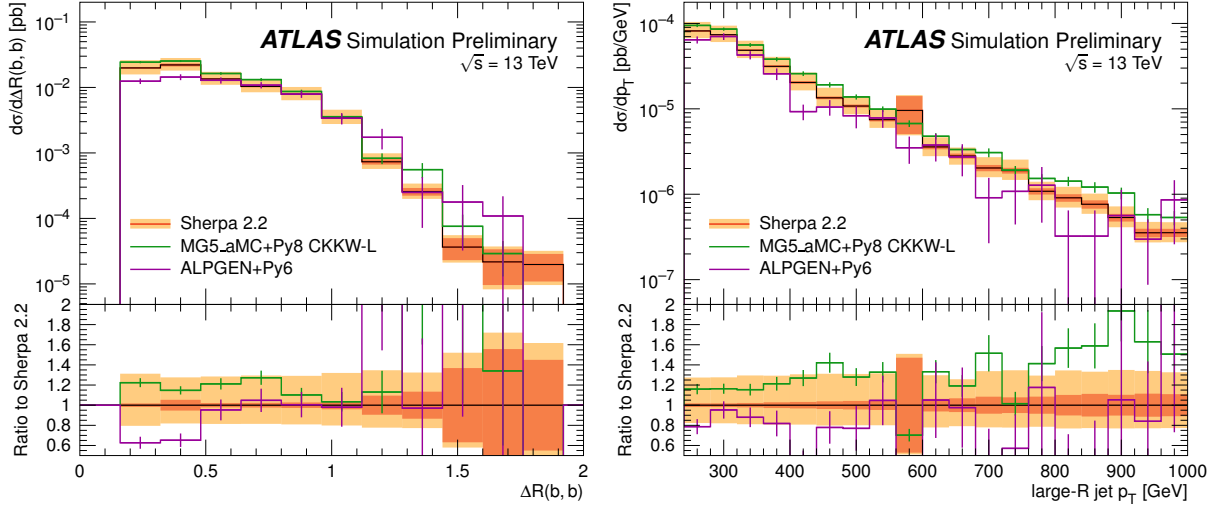


Figure 18: Predictions for the differential cross sections as a function of the rapidity difference between the two b -tagged jets (left) and the transverse momentum of the large- R jet in events with two b -tagged jets from SHERPA 2.2 (red), MG5_aMC@NLO+Py8 using CKKW-L (blue) and ALPGEN+PYTHIA6 (purple). The orange band includes PDF and scale uncertainties estimated with SHERPA 2.2, while the size of the statistical uncertainty components are indicated by the size of the error bars.

exceeding the other two predictions. Shapes of the jet mass of the large- R jet are very similar for the two LO generators, but differ at low values drastically for SHERPA. The three predictions are in very good agreement shape-wise, and within uncertainty for the normalisation, for the transverse momentum of the large- R jet.

In the b -tagged distribution, the statistics available from the three predictions is rather low. Nonetheless, meaningful comparisons in the $\ln \Delta R_{bb}$ region, of particular interest, because dominated by gluon splitting to a $b\bar{b}$ pair, and in most of the analysed range of the large- R jet transverse momentum can be carried out. While the predictions for low ΔR_{bb} are compatible within uncertainties between MG5_aMC+Py8 CKKW-L and SHERPA, the ALPGEN+PYTHIA6 calculation shows a different shape, with a clear slope in the first four bins of the analysed distribution. The p_T prediction is compatible among the three generators within the large uncertainties.

5.3 Relevant variables for the $VH(H \rightarrow b\bar{b})$ analysis

As mentioned above, the $V + b(b)$ process is one of the main irreducible backgrounds for in the $VH(H \rightarrow b\bar{b})$ search. Moreover, in such a search it is often difficult to identify a pure enough control region where constrains or information on this background process could be extracted. For this reason, the knowledge of the modelling of the $V + b(b)$ processes is crucial, and the uncertainty coming from the predictions is one of the largest affecting the Higgs signal extraction, as can be observed in the latest ATLAS results on $VH(H \rightarrow b\bar{b})$ [46]. In the following, studies aimed at understanding the impact of the $V + b(b)$ modelling in the specific VH analysis-regions are reported.

In the analysis, jets are reconstructed using the anti- k_t clustering algorithm with a jet-radius parameter of $R = 0.4$. The jet transverse momentum is required to be greater than 20 GeV for central jets ($|\eta| < 2.5$) and

greater than 30 GeV for forward jets ($2.5 < |\eta| < 5$), and the leading jet is required to have a transverse momentum larger than 45 GeV. Selected central jets are labelled according to their flavour with the following procedure:

1. if a b -hadron is found within a cone of $\Delta R = 0.3$ centered along the jet axis (considering all particles that belong to the jet constituents and their ancestors in the MC truth record) the jet is b -labelled;
2. if a jet is not b -labelled, and a c -hadron is found within a cone of $\Delta R = 0.3$ centered along the jet axis the jet is c -labelled;
3. if a jet is not b -labelled neither c -labelled, it is then labelled as light-flavour (l -labelled).

To select events in which the vector boson is produced in association with heavy flavor jets, only events where the 2 selected central jets are labelled as bb , bc , bl , cc are retained, and defined as $V + hf$. Therefore, the phase space of interest for this study is different from what shown in sections 5.1–5.2, where only b -jets were considered in the final state. The missing transverse energy is reconstructed by taking the transverse momentum of the negative sum of the four-momenta of all the visible particles, and the missing transverse momentum is defined consistently summing over all the charged particles.

Three event selections are implemented, corresponding to the relevant three search channels: the 0, 1 and 2-lepton channel selections, corresponding to the $Z \rightarrow \nu\nu$, the $W \rightarrow \ell\nu$, and the $Z \rightarrow \ell\ell$ channels. All selections require at least 2 central jets, and for the $Z \rightarrow \nu\nu$ and $W \rightarrow \ell\nu$ ones a veto on a higher number of jets (central + forward) than 3 is also applied. Further details relative to the specific channels are listed in the following, where plots of the relevant variables studying the effect of scale variations on SHERPA 2.2, varying μ_F and μ_R of a factor of two up or down in different combinations, and comparing the result with MG5_aMC+Py8 CKKW-L B (for the $Z \rightarrow \ell\ell$ and $W \rightarrow \ell\nu$ channels) or MG5_aMC+Py8 CKKW-L A (for the $Z \rightarrow \nu\nu$ channel) are shown. Only shape effects are taken into account for this study, therefore the distributions are all normalised to the same area.

5.3.1 $Z \rightarrow \nu\nu$ channel

The $Z \rightarrow \nu\nu$ -channel particle-level selection is defined as follow:

- $E_T^{\text{miss}} > 150$ GeV, and no charged leptons in the final state
- $\Delta\phi(E_T^{\text{miss}}, p_T^{\text{miss}}) < \pi/2$
- $\min[\Delta\phi(E_T^{\text{miss}}, jets)] < 20^\circ$ (using the first 3 leading jets)
- $\sum p_T(jets) > 120(150)$ GeV for 2(> 2)-jet events (the sum is over the first 2 leading central jets and the first additional jet in p_T)
- $\Delta\phi(j_1, j_2) < 140^\circ$
- $\Delta\phi((E_T^{\text{miss}}, j_1 + j_2) < 120^\circ$

The variables which are affected more by the effect of scale variations and where the shape comparison between SHERPA and MG5_aMC+Py8 CKKW-L are more pronounced are shown in figures 19–20.

Despite the effect of the scales can be appreciable on some of these distributions, the largest difference in the di- b -jet variables shown in figure 19 arises from the comparison of the two generators, which both use a 5FNS, but a different order of the prediction, NLO for SHERPA, and LO for MG5_aMC+Py8 CKKW-L.

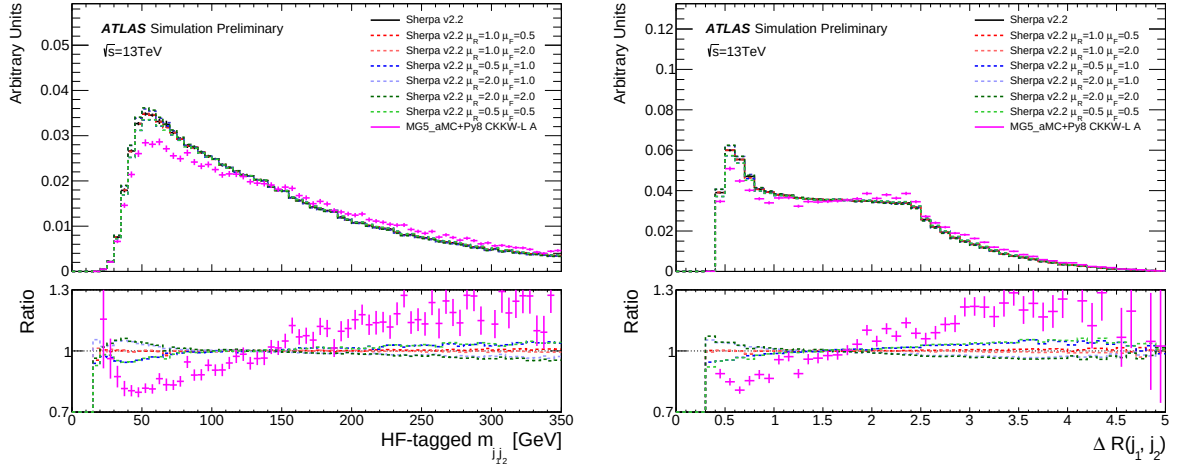


Figure 19: $Z \rightarrow \nu\nu$ -channel: Shape comparisons of the distribution of the invariant mass of the b -jet pair (left), and of the angular separation between the two b -jets (right) in events satisfying the selection for a $VH(H \rightarrow b\bar{b})$ signal. The SHERPA 2.2 curve is used as nominal prediction, and compared to the curves obtained varying the μ_F and μ_R scales, and to the MG5_aMC+Py8 CKKW-L prediction. The ratio is performed with respect to the SHERPA 2.2 nominal curve.

In the case of the N_{jet} and $\Delta\phi$ distributions, the two effects are comparable, and the $p_T(V)$ behavior is the opposite, with madgraph over predicting the high- p_T region with respect to SHERPA.

5.3.2 W channel

The $W \rightarrow \ell\nu$ -channel particle-level selection is defined as follow:

- exactly 1 lepton with $p_T > 25$ GeV and $|\eta| < 2.47$ (for the electron case) 2.7 (for the muon case)
- $p_T^W > 150$ GeV

The variables which are affected more by the effect of scale variations and where the shape comparison between SHERPA and MG5_aMC+Py8 CKKW-L are more pronounced are shown in Figure 21. The W channel is probably the most crucial of the three channels, because of the difficulty to find a control region sufficiently pure in $W + b\bar{b}$ events. Therefore the estimate of such background strongly relies on MC predictions.

Also in this case, the difference between MG5_aMC+Py8 CKKW-L and SHERPA is larger than the SHERPA scale variations, and the plots show a similar trend than in the case of the $Z \rightarrow \nu\nu$ channel, shown in Figures 19–20.

5.3.3 $Z \rightarrow \ell\ell$ channel

The $Z \rightarrow \ell\ell$ -channel particle-level selection is defined as follow:

- exactly 2 same-flavour (and opposite charge for muons) satisfying the criteria: $p_T > 25$ GeV and $|\eta| < 2.47$ (for electrons) 2.7 (for muons)

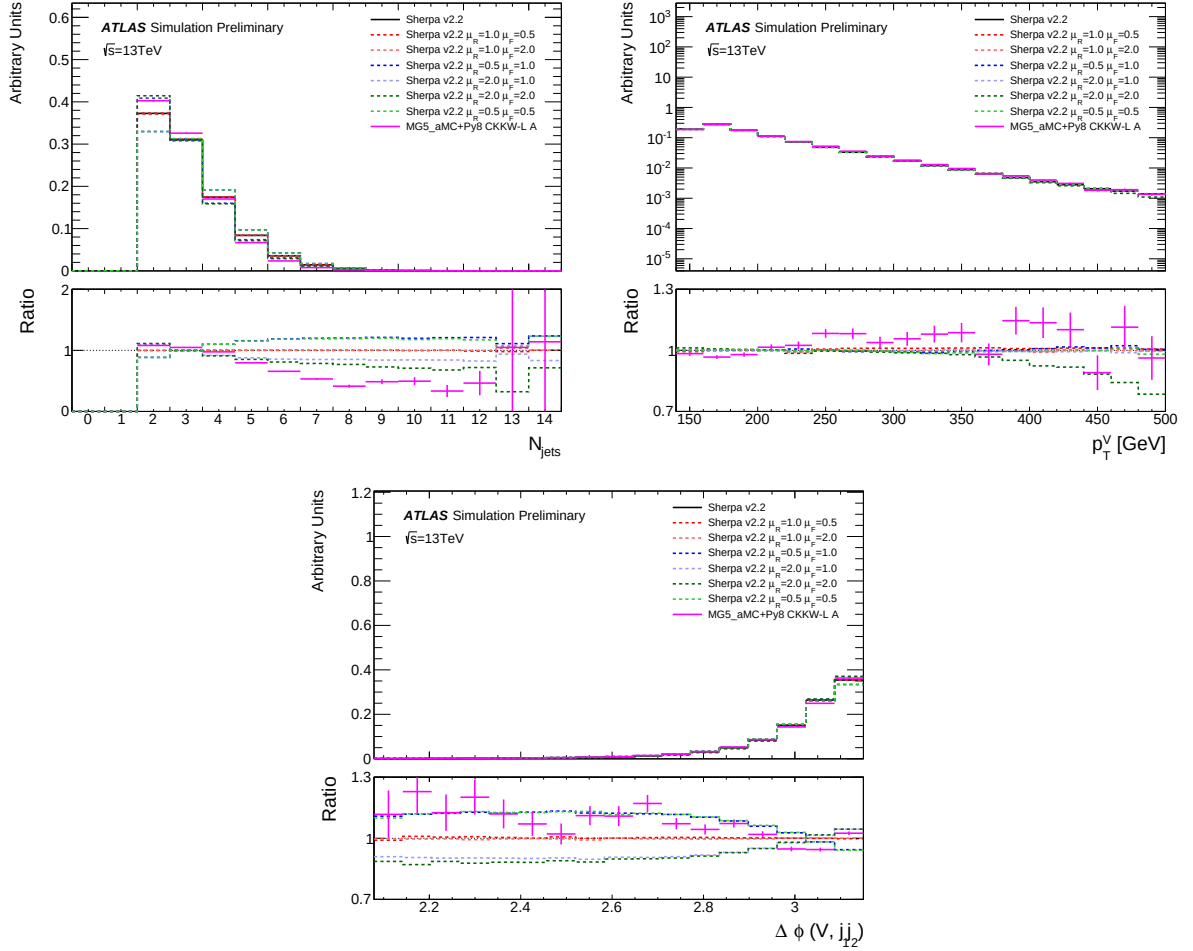
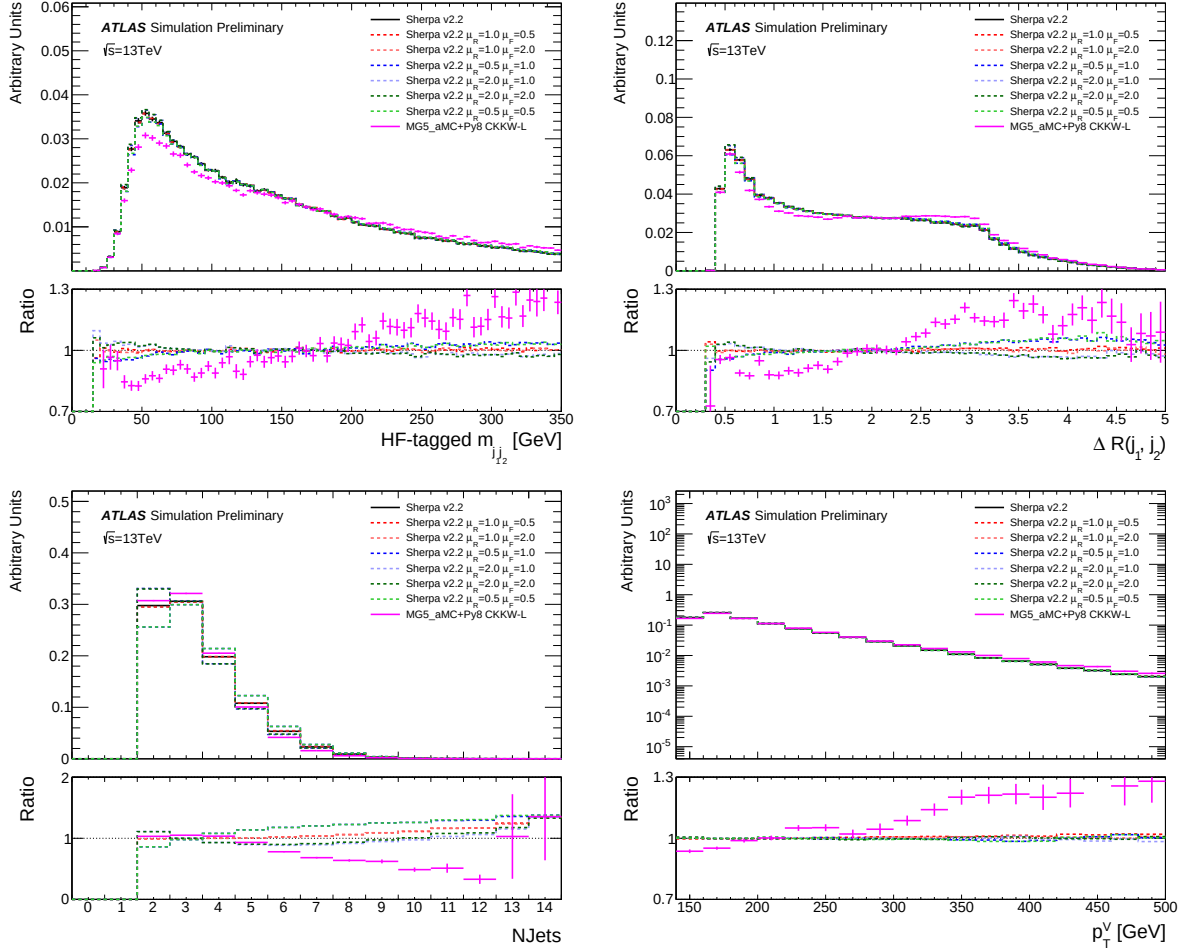


Figure 20: $Z \rightarrow \nu\nu$ -channel: Shape comparisons of the distribution of the number of jets (top left), of the vector boson transverse momentum (top right) and of the azimuthal separation between the vector boson and the $b\bar{b}$ system (bottom), in events satisfying the selection for a $VH(H \rightarrow b\bar{b})$ signal, except the requirement on the maximum number of jets, relaxed for the jet multiplicity plot. The SHERPA 2.2 curve is used as nominal prediction, and compared to the curves obtained varying the μ_F and μ_R scales, and to the MG5_aMC+Py8 CKKW-L prediction. The ratio is performed with respect to the SHERPA 2.2 nominal curve.



- $71 \text{ GeV} < m_{\ell\ell} < 121 \text{ GeV}$

The variables which are affected more by the effect of scale variations and where the shape comparison between SHERPA and MG5_aMC+Py8 CKKW-L are more pronounced are shown in Figure 22.

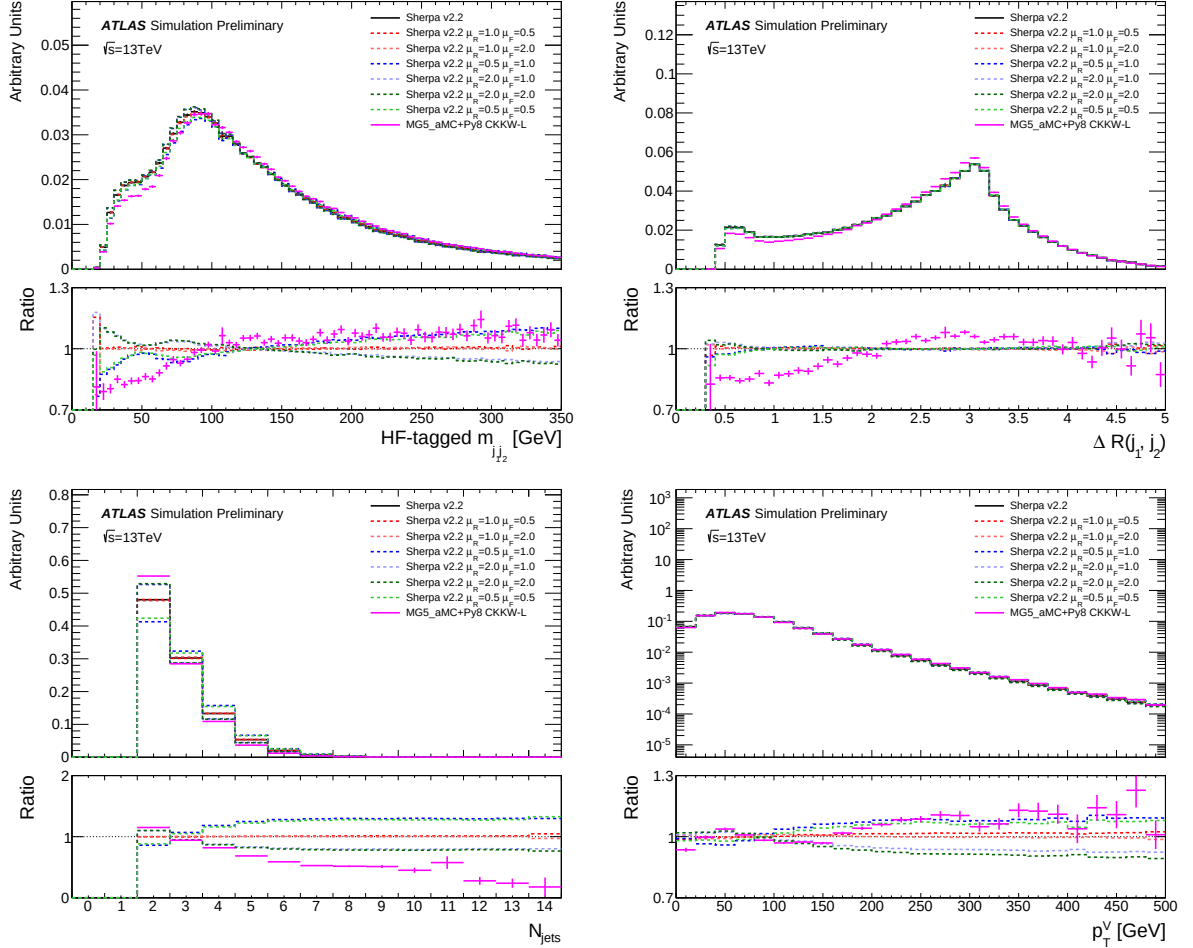


Figure 22: $Z \rightarrow \ell\ell$ -channel: Shape comparisons of the distribution of the invariant mass of the di- b -jet pair (top left), of the angular separation between the two b -jets (top right), of the jet multiplicity (bottom left), and of the transverse momentum of the vector boson (bottom right), in events satisfying the selection for a $VH(H \rightarrow b\bar{b})$ signal. The SHERPA 2.2 curve is used as nominal prediction, and compared to the curves obtained varying the μ_F and μ_R scales, and to the MG5_aMC+Py8 CKKW-L prediction. The ratio is performed with respect to the SHERPA 2.2 nominal curve.

This case is similar presents similar trends to the one shown before, and more consistency between the scale variations of SHERPA, and the difference to MG5_aMC+Py8 CKKW-L. This is coherent to what observed in the $Z \rightarrow \nu\nu$ channel, as expected.

6 Electroweak Vjj production

The V + jets final state can also be constructed with two additional electroweak couplings at tree level, giving rise to a gauge-invariant set of Vjj Feynman diagrams, in which the two jets recoil against a t -channel gauge boson at tree level. This includes the vector boson fusion diagram. Vjj predictions have been generated using SHERPA 2.2 with up to two extra emissions at LO accuracy and are shown here together with an estimate of the scale and PDF uncertainties to illustrate the typical size of the generator uncertainties in regions of phase space probed by ATLAS electroweak Vjj analyses. The samples only have LO accuracy and do not include any higher-order QCD or electroweak corrections.

6.1 Electroweak Wjj production

Comparisons of 13 TeV predictions for electroweak Wjj production are shown for two regions of phase space, that have recently been measured by ATLAS at 8 TeV [43]. The Wjj *inclusive* region requires at least two jets with $p_T > 80$ GeV and 60 GeV, respectively. Jets are reconstructed using the anti- k_t algorithm using a jet-radius parameter of 0.4. The rapidity separation between the two jets is required to be greater than $\Delta y > 2$ and the invariant mass of the dijet system, m_{jj} is required to be greater than 500 GeV. Furthermore, a dressed leptons with $p_T > 25$ GeV and $|\eta| < 2.5$ is selected and the missing transverse energy, E_T^{miss} , is required to be greater than 20 GeV. The transverse mass given by

$$m_T = \sqrt{2p_T \cdot E_T^{\text{miss}} [1 - \cos \Delta\phi(\ell, E_T^{\text{miss}})]} \quad (2)$$

is required to be 40 GeV and the distance between the lepton-jet distance, $\Delta R = ((\Delta y)^2 + (\Delta\phi)^2)^{1/2}$, has to be greater than 0.3.

Figure 23 shows the normalised distributions of the lepton centrality and the leading jet transverse momentum in the inclusive region. The dijet invariant mass spectrum as well as the dijet rapidity difference are shown in the inclusive regions in Figure 24. The generator uncertainties have been estimated using SHERPA 2.2 and are indicated by the orange band in the ratio panels of the plots. The band is constructed from the statistical uncertainty of the SHERPA predictions, the PDF uncertainties from the NNPDF3.0nnlo replicas estimated using LHAPDF along with α_S variations of ± 0.002 around the nominal value of 0.118 as well as an envelope formed from 7-point scale variations of the renormalisation and factorisation scales. The various uncertainty components are added in quadrature to give the total uncertainty shown in the plots.

The Wjj *signal* region is constructed from the subset of events that pass the inclusive region and also pass $m_{jj} > 1$ TeV. Exactly one central lepton and no central jets are required in the signal region, where the number of central leptons is determined from counting the number of leptons that have a lepton centrality $C_\ell < 0.4$ where

$$C_\ell = \left| \frac{y_\ell - \frac{y_1 + y_2}{2}}{y_1 - y_2} \right| \quad (3)$$

corresponds to the rapidity location of the lepton relative to the average rapidity of the two leading jets. For the jet veto, a jet centrality is defined in a similar way for additional jets in the event. Figure 25 shows the normalised distributions of the dijet invariant mass spectrum and the dijet rapidity difference in the signal region.

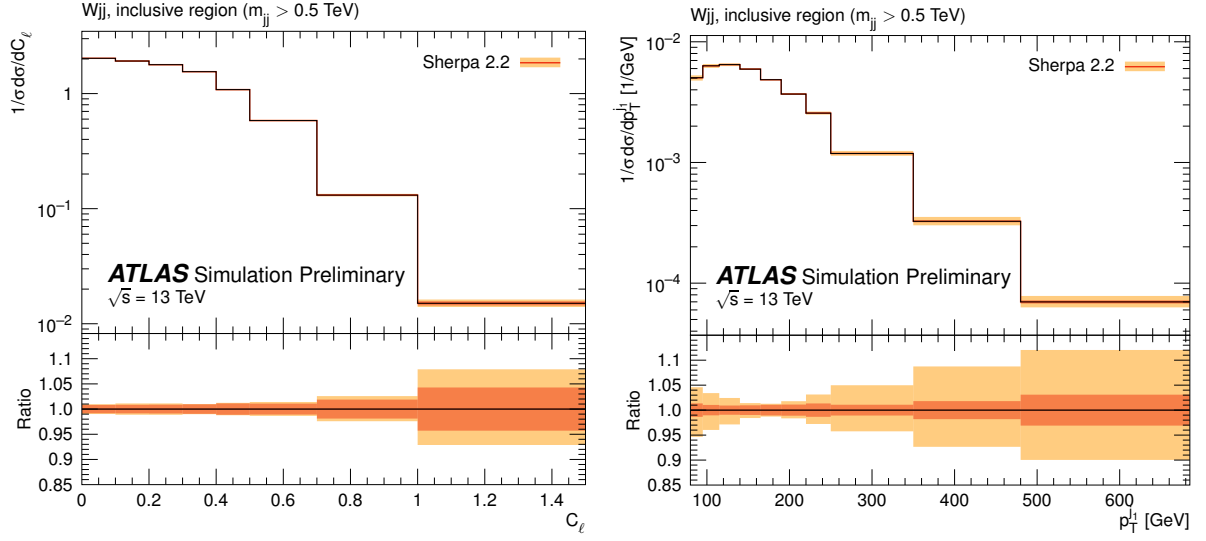


Figure 23: Normalised prediction for the lepton centrality (left) and the leading jet p_T (right) from SHERPA 2.2 in the inclusive phase space. The yellow band includes statistical, PDF and scale uncertainties, while the red error bars indicate the size of the statistical component.

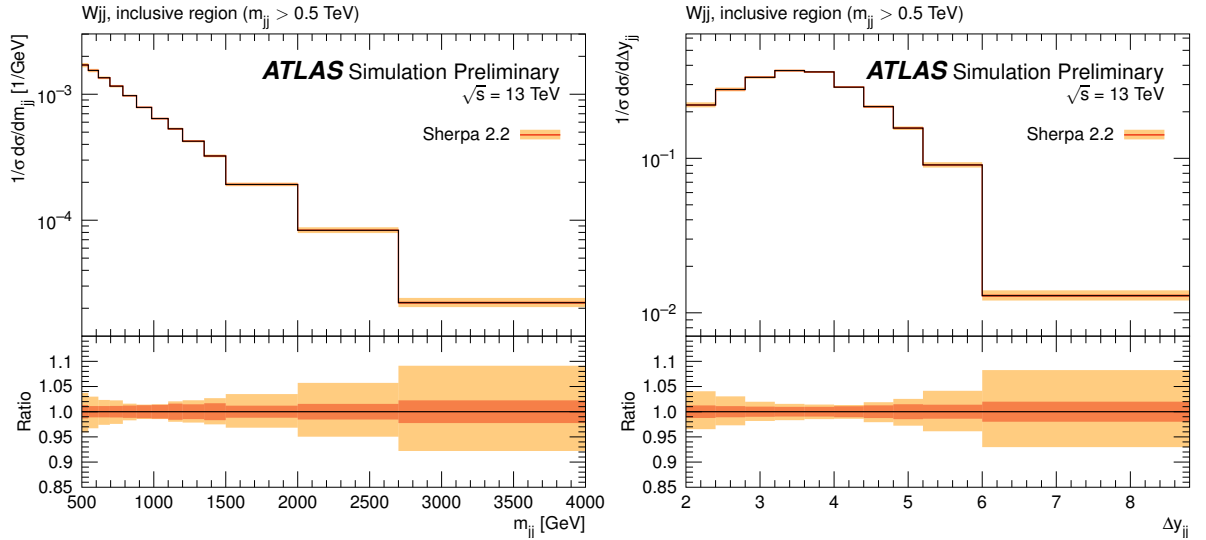


Figure 24: Normalised prediction for the invariant dijet mass (left) and the rapidity difference of the dijet system (right) from SHERPA 2.2 in the inclusive phase space. The yellow band includes statistical, PDF and scale uncertainties, while the red error bars indicate the size of the statistical component.

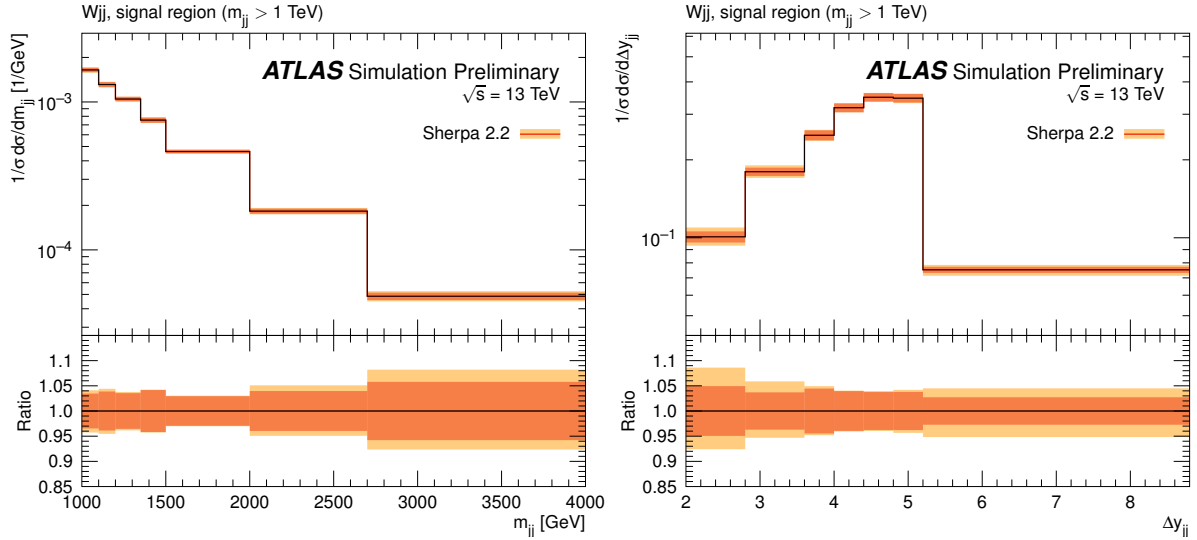


Figure 25: Normalised prediction for the invariant dijet mass (left) and the rapidity difference of the dijet system (right) from SHERPA 2.2 in the signal region with $m_{jj} > 1$ TeV. The yellow band includes statistical, PDF and scale uncertainties, while the red error bars indicate the size of the statistical component.

6.2 Electroweak Zjj production

Comparisons of 13 TeV predictions for electroweak Zjj production are shown for two regions of phase space, that have been measured by ATLAS at 8 TeV [47]. The Zjj *inclusive* region requires at least two jets with $p_T > 55$ GeV and 45 GeV, respectively. Jets are reconstructed using the anti- k_t algorithm using a jet-radius parameter of 0.4. Furthermore, a Z-boson candidate is constructed from exactly two opposite-sign same-flavour dressed leptons with $p_T > 25$ GeV and $|\eta| < 2.47$ and the invariant mass of the dilepton system is required to fall within ± 10 GeV of the resonance value.

Figure 26 shows the predicted differential cross sections as a function of the jet multiplicity and the leading jet transverse momentum in the inclusive phase space. Scale and PDF uncertainties have been estimated in the same way as described in Section 6.1 and are indicated by the orange band.

The Zjj *search* region is made up from the subset of events that pass the inclusive region as well as a veto on additional jet-activity in between the two leading jets in order to suppress regular $Z +$ jets production. A veto scale of 20 GeV is used for additional jets beyond the two leading jets. The transverse momentum of the dilepton system is required to be greater than 20 GeV and the vector p_T sum of the two leptons and the two jets, which is normalised by the scalar p_T sum, is required to be less than 0.15. Finally, the dijet invariant mass, m_{jj} , is required to be greater than 1 TeV.

Figure 27 shows the predicted differential cross sections as a function of the dijet invariant mass and the dijet rapidity difference. The predicted differential cross section as a function of the dilepton p_T spectrum is shown in Figure 28. The uncertainties are at the level of 10–20 % in the inclusive Zjj regions, but can rise beyond 30 % in the high-mass search region, which has a 1 TeV cut on the dijet invariant mass and a veto on additional jet activity in the rapidity interval between the two jets in order to suppress backgrounds, including regular Drell–Yan production.

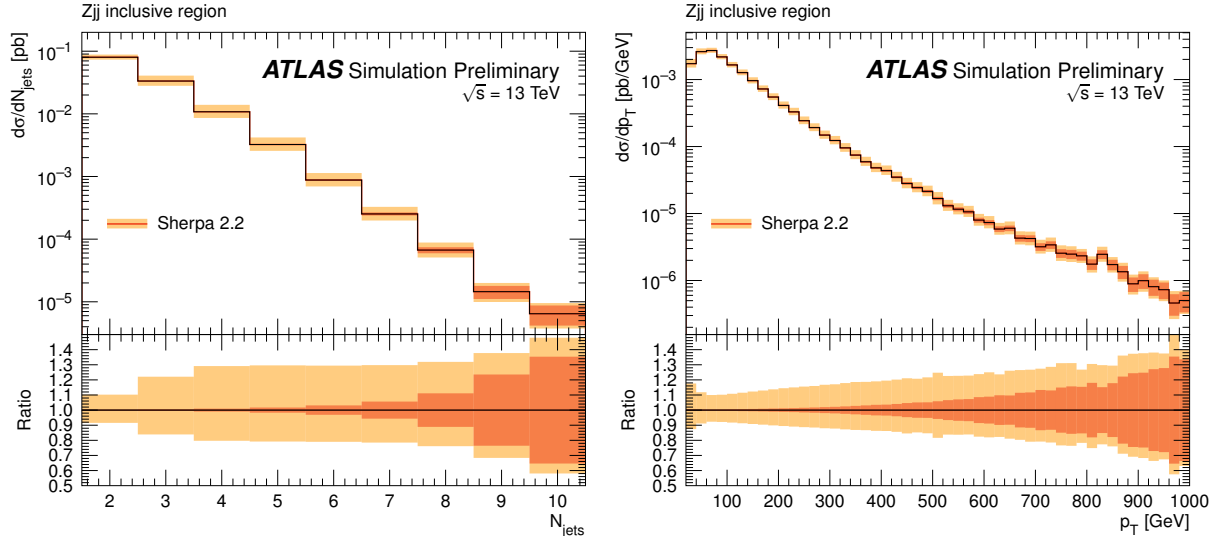


Figure 26: Prediction for the differential cross sections as a function of the jet multiplicity (left) and the leading jet p_T (right) from SHERPA 2.2 in the inclusive region with $m_{jj} > 0.5$ TeV. The yellow band includes statistical, PDF and scale uncertainties, while the red error bars indicate the size of the statistical component.

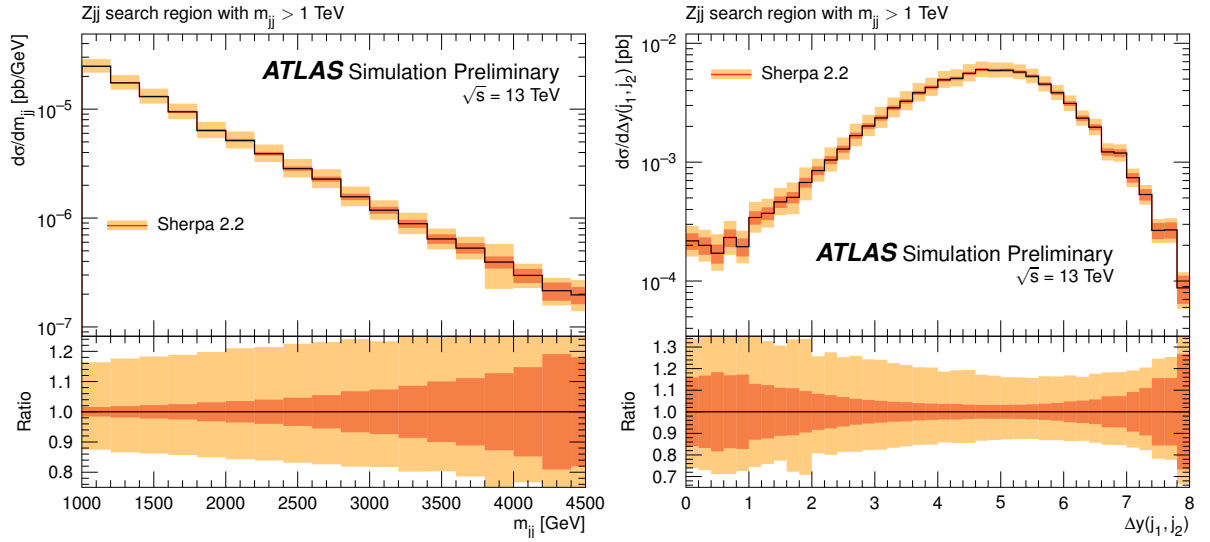


Figure 27: Prediction for the differential cross sections as a function of the invariant dijet mass (left) and the rapidity difference between the two leading jets (right) from SHERPA 2.2 in the search region with $m_{jj} > 1$ TeV. The yellow band includes statistical, PDF and scale uncertainties, while the red error bars indicate the size of the statistical component.

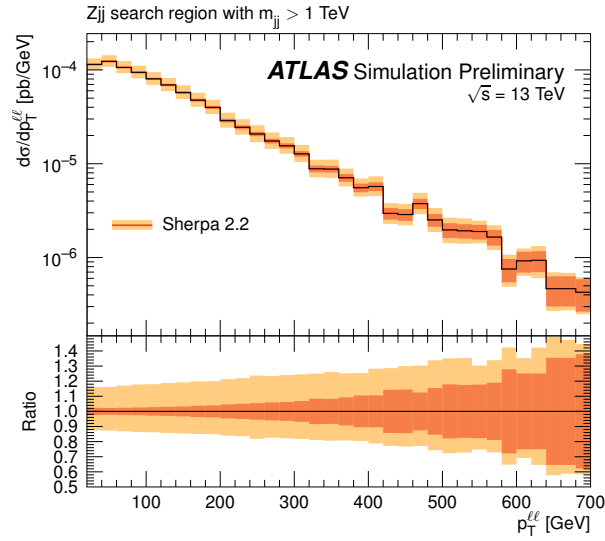


Figure 28: Prediction for the differential cross sections as a function of the dilepton transverse momentum from SHERPA 2.2 in the search region with $m_{jj} > 1$ TeV. The yellow band includes statistical, PDF and scale uncertainties, while the red error bars indicate the size of the statistical component.

7 Tuning PYTHIA8 parameters in matched setups with MG5_aMC+Py8 CKKW-L

7.1 Motivation

The PYTHIA8 A14 tune used in ATLAS had the parameter `SpaceShower:rapidityOrder` turned on. This forces the initial state radiations (ISR), after the first, to be ordered in rapidity. However, when showering events generated by MG5_aMC+Py8 CKKW-L, this setting results in non-negligible dependence on the merging scale [48]. So it was decided to start from the A14 tune, and after turning this particular setting off, re-tune PYTHIA8 parameters in Z-boson+jets events. The A14 tune was performed for four different PDFs, but in ATLAS the version with NNPDF2.3LO is used, and in this note, A14 refers to this particular variant only.

7.2 Strategies

1. In the first approach, referred to as Tune (RO), in MG5_aMC+Py8 CKKW-L, Z-boson events with up to 3 extra jets were generated, and showered with PYTHIA8 A14 tune. Then the tuning was performed on the complete matched setup, *i.e.* on the output of PYTHIA8 interfaced with MG5_aMC+Py8 CKKW-L.
2. In the second approach, referred to as PS-only tune (RO), it was decided to initially decouple the two generators. The tuning of PYTHIA8 was done separately using only PYTHIA8 output. Then the obtained tunes were used to shower Z-boson events with up to 3 extra jets generated in MG5_aMC+Py8 CKKW-L, and the consistency with initial PYTHIA-eight only tune predictions were checked.

In both cases the Rivet analysis toolkit and the PROFESSOR MC tuning system have been used, at versions 2.5.3 and 2.1.4 respectively. The PYTHIA version used was 8.219, with the PDFs taken from LHAPDF version 6.1.3. The MG5_aMC+Py8 CKKW-L parameters used were: `Merging:Dparameter=0.2`, `Merging:unorderedASscalePrescrip=0`, and `Merging:TMS=30`.

In ATLAS, tunes are performed using the PROFESSOR automated tuning tool, where each bin of each observable is parametrised as a N -dimensional 3rd order polynomial based on the sampled tune points in the parameter hypercube (N being the number of tuned parameters). These parametrisations are then used to calculate a χ^2 with respect to the reference data, which was numerically minimised in the parameter space to find the best tune point. Weight factors are used in the χ^2 and degree of freedom, N_{df} , calculation to place increased emphasis on certain observables.

The main focus in each case was to describe the distributions sensitive to underlying event in Z-boson events [49], so most weight was put on that. However, other analyses like measurement of event-shape in Z-boson events [50] was also looked at, and care was taken so that the Z-boson p_T distribution [51] was modelled well in each case. These three analyses were performed at $\sqrt{s} = 7$ TeV.

In the first approach, the α_S corresponding to ISR, FSR and MPI were set to 0.125, motivated by [52]. Then the multiple parton interaction (MPI) parameter corresponding to p_T cutoff, `MultipartonInteractions:pT0Ref` and the colour reconnection strength parameter `ColourReconnection:range` were tuned.

In the second approach, more PYTHIA8 parameters were tuned. They divide roughly into the initial and final state parton shower scales and couplings (**SpaceShower:*** and **TimeShower:*** parameters), and multiple parton interactions (**MultipartonInteractions:*** parameters). Two “beam-remnant” parameters are also tuned: the **primordialKtHard** which generates a narrow Gaussian intrinsic transverse momentum for the hard process incoming partons, and the previously mentioned **ColourReconnection:range**.

Traditionally, the distribution of mean p_T against charged particle multiplicity is found to be very sensitive to the modelling of the colour reconnection (CR) in underlying event analyses. The PYTHIA8 model for CR has recently undergone a large expansion in [53], and it was decided to test the effect of these new models as well while tuning. While all the current tunes uses the so called “MPI-based” scheme (will be referred to as CR0 model subsequently, where the **ColourReconnection:range** is the tunable parameter), two new models of CRs have been introduced, with corresponding tunable parameters. In the first new one, which adds more degrees of freedom to the MPI-based scheme (will be referred to as CR1 model subsequently), two parameters, **ColourReconnection:m0** corresponding to the string length and **ColourReconnection:junctionCorrection** are tuned. It was used with **ColourReconnection:allowDoubleJunRem** = **off** setting, as recommended by the authors. In the second new model, termed the “gluon-move scheme” (will be referred to as CR2 model subsequently), three parameters were tuned: **ColourReconnection:m2Lambda**, **ColourReconnection:fracGluon**, and **ColourReconnection:dLambdaCut**.

A detailed description of the tuned parameters is beyond the scope of this note, and the reader is referred to the [PYTHIA8 online documentation](#).

7.3 Results and discussions

The table 3 documents the result for each of the tuning setups mentioned earlier. In each case, the parameters not mentioned were fixed at A14 value.

In Fig. 29, the performance of these tunes is shown against different ATLAS measurements. In general, the best possible tune obtained for each setup give very similar results. The plots of UE in Z-boson events are most sensitive to MPI. The charged particle sum p_T and multiplicity density as a function of Z-boson p_T in toward region show similar trends. Just switching ISR rapidity ordering off clearly deteriorates the agreement with data considerably, necessitating this tuning effort. For mean p_T against charged particle multiplicity, none of the models predict the right shape, in spite of they having different CR models. This is in strong contrast to what happens in minimum-bias events [54]. For the thrust distribution at low Z-boson p_T , where it should be most sensitive to MPI, the trend is somewhat reversed, with new tunes doing a bit worse than the A14 with ISR rapidity ordering off. Most of the tunes, except PS-only tune (RO, CR2) do very well for Z-boson p_T , which is affected mostly by ISR and BBR intrinsic k_T . Finally, for inclusive jet multiplicity in Z-boson events most tune predictions are very similar, but they all underestimate the data.

In Fig. 30, the effect of varying the merging scale by a factor of two and half on the UE observables in Z-boson events for PS-only tune (RO, CR0) tune is shown. This confirms little or no dependence on the merging scale even for the tunes where only PYTHIA8 output is tuned.

The following points discuss the interesting features and open issues:

- The robustness of the tunes can be clearly be seen, as tunes obtained by two different approaches give very similar predictions.

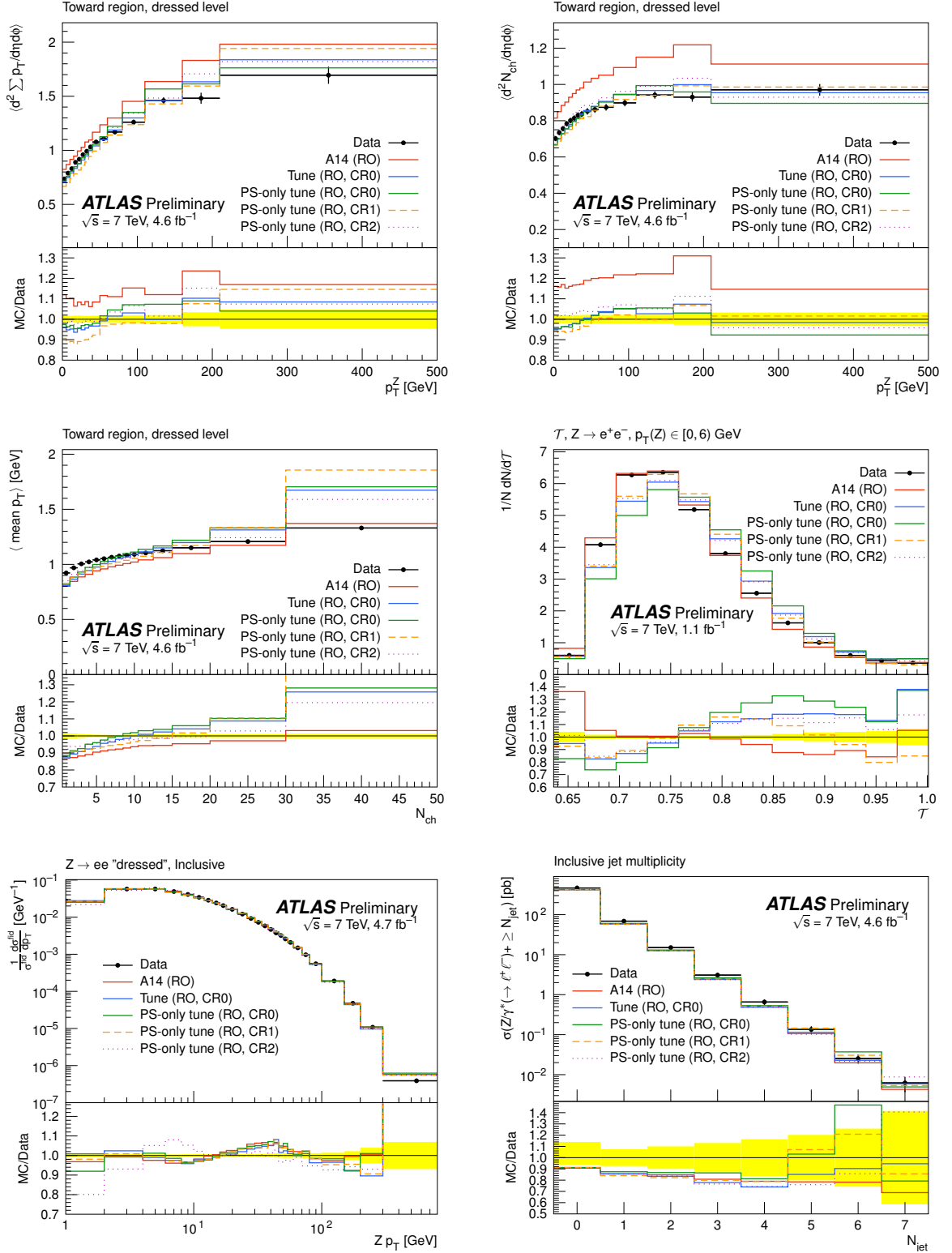


Figure 29: Predictions from the obtained tunes, and from A14 with ISR rapidity ordering off, compared with ATLAS Z-boson underlying event [49], Z-boson event shape [50], Z-boson p_T and jet multiplicity [51] in Z-boson events observables. The yellow shaded areas represent data uncertainty.

Parameter	A14/ Default (range)	PS-only tune			Tune
		CR0	CR1	CR2	
MultipartonInteractions:pT0Ref	2.09	2.05	1.89	1.97	2.20
MultipartonInteractions:expPow	1.85	1.94	2.97	2.47	NT
MultipartonInteractions:alphaSvalue	0.126	NT	NT	NT	0.125
BeamRemnants:primordialKThard	1.88	2.24	1.86	2.73	NT
ColourReconnection:range	1.71	3.35	NA	NA	6.5
ColourReconnection:m0	0.3 (0.1 - 5)	NA	3.23	NA	NA
ColourReconnection:junctionCorrection	1.20 (0.01 - 10)	NA	3.31	NA	NA
ColourReconnection:m2Lambda	1.0 (0.25-16)	NA	NA	8.87	NA
ColourReconnection:fracGluon	1.0 (0-1)	NA	NA	0.93	NA
ColourReconnection:dLambdaCut	0 (0-10)	NA	NA	0.90	NA
SpaceShower:alphaSvalue	0.125	0.127	0.125	0.121	0.125
SpaceShower:pT0Ref	1.30	2.35	1.40	1.58	NT
SpaceShower:pTdampFudge	1.21	0.80	0.92	1.09	NT
TimeShower:alphaSvalue	0.126	0.132	0.142	0.118	0.125
TimeShower:pTmin	0.50	1.45	1.51	1.52	NT
TimeShower:pTdampFudge	1.0	0.81	1.18	1.34	NT

Table 3: Obtained tuned values for different parameters for each setup is shown. NT and NA respectively denote “not tuned” and “not applicable”. The first case is when that particular tune kept the default A14 value of that parameter, and in the latter case, that parameter is not applicable for that particular setup. In the first column, for CR1 and CR2 settings, rather than showing NA (for A14), the default values suggested by the authors, with the allowed range is shown. The values in *italics* indicate that they were fixed during the tuning process.

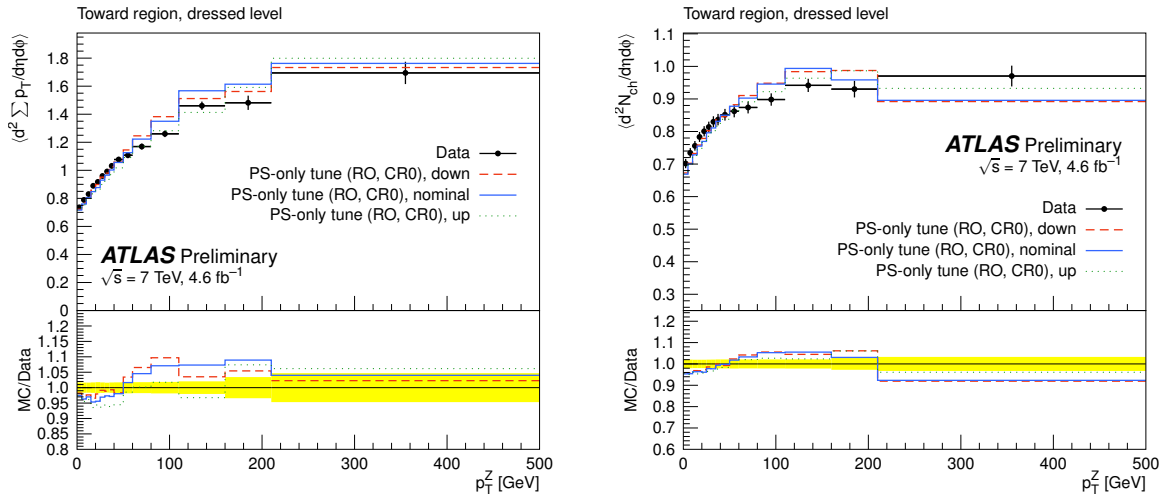


Figure 30: Variation in prediction of PS-only tune (RO, C02) tune, when merging scale is set to 15 GeV (down) and 60 GeV (up), from the default 30 GeV (nominal). The other distributions show even smaller or negligible change. The yellow shaded areas represent data uncertainty.

- One of the interesting features observed in Tune (R0, CR0) was the relatively high value of CR strength parameter. While the model allows values up to 10, for all previous PYTHIA8 tunes, this had a lower value, between 1 to 2. However, it can be seen in PS-only tune (RO, CR0) tune, that a somewhat lower value that Tune (RO, CR0) can give similar level of agreement, when additional parameters are also tuned. But even then there seemed like is a preference for higher value than previous tunes, so it will be interesting to understand if this is due to the matched setup with MG5_aMC+Py8 CKKW-L, or due to turning ISR rapidity ordering off.
- It should also be noted that in spite of trying the newer CR models, the shape of the mean p_T against multiplicity distribution could not be improved, when other UE observables and p_T of the Z-boson were also required to modelled well. So that points to some tension between these observables, which does not seem to be the case in minimum-bias events.

8 Conclusion

The baseline Monte Carlo simulation setup used by ATLAS to model single electroweak boson production in association with jets in proton–proton collisions at a centre-of-mass energy of 13 TeV in Run 2 has been described. Several Monte Carlo generator predictions are compared in regions of phase space relevant for Run 2 data analyses, but also compared to recent measurements using Run-1 or early Run-2 data that has been corrected for detector effects. Comparisons are made for regions of phase space with or without additional requirements on the heavy-flavour content in the accompanying jets, as well as electroweak Vjj production. Large discrepancies between different predictions are highlighted for processes involving heavy-flavour jets and PDF and scale uncertainties are studied for all processes. In addition, dedicated sets of tuned PYTHIA8 parameters in matched setups with MG5_aMC+Py8 CKKW-L are presented and discussed.

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